



MediaLedger

Version 2.4.2

October 2026

User Manual

Inventory, review and tidy a Plex media share.
For the Windows app, the Raspberry Pi server
and the family portal.

Windows

Raspberry Pi

Family portal



How to use this manual

MediaLedger keeps a searchable ledger of your TV, anime and movie files, shows what is missing or worth upgrading, and can rename files safely. This manual is arranged by what you are trying to do, not by menu.

I am new Read Part 1 in order. It takes about ten minutes and ends with your first scan. Start at chapter 1 , then install it .	I use it daily Jump to the page you are on. Every chapter opens with a strip: where it is in the sidebar, who can open it, and the question it answers. Start with the Dashboard and Missing .	I run the server Part 7 covers the Raspberry Pi install, shares, updates, HTTPS and publishing. Start with the Raspberry Pi chapter , then Security .
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Conventions in this manual

You see	It means
Strip under a chapter title	Where to open it (WHERE), which roles can (WHO CAN OPEN IT), and the question it answers (IT ANSWERS).
NOTE (blue bar)	Background that helps, with no action needed.
TIP (green bar)	A shortcut or a better way to do it.
CAUTION (amber bar)	Something that can surprise you, cost time or hide data.
IMPORTANT (red bar)	Something that can lose data or expose your library. Read it before you continue.
Dark block with green text	A command or file to type or paste exactly. Lines starting with <code>sudo</code> run on the Raspberry Pi.
Blue underlined text	A link to another section. In the PDF and Word file, click it to jump there.
Figure N	A screenshot or diagram, numbered through the whole manual.

Find it fast

Look up the job, then follow the link. Page numbers are in the second column; every reference in the manual is clickable.

I want to...	Go to
Install MediaLedger and run the first scan	chapter 2, p. 12 and section 2.3, p. 12
Change what the Dashboard shows or its colours	section 4.7, p. 20
See which episodes I am missing, and tell it which ones I do not want	chapter 9, p. 31
Correct a wrong match or a mis-parsed title	section 10.2, p. 34 and section 9.3, p. 32
Find duplicates and files in the wrong place	section 10.3, p. 35
Find low-quality files worth replacing	chapter 11, p. 36
Free up disk space safely	section 13.2, p. 42
Rename movies or episodes, and undo a rename	chapter 15, p. 45 and section 15.6, p. 48
Let family browse and request titles	chapter 19, p. 56
Get a message when something happens	chapter 20, p. 59
Link Plex for ratings and watch history	chapter 23, p. 68
Back up and restore the database	section 21.4, p. 62
Set up the Raspberry Pi server	chapter 26, p. 79
Reach it by a name, or from outside the house	section 27.1, p. 85 and section 27.4, p. 87
Lock it down: roles, two-factor, HTTPS	chapter 25, p. 74
Fix something that is not working	chapter 28, p. 90
Look up a default, an address or a command	chapter 29, p. 93

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PART 1

Start here

What MediaLedger is, how to install it on Windows and how to find your way around.

IN THIS PART

- 1 What MediaLedger is**..... 8
The idea, the three ways to run it, what it never does, and how a scan works.
- 2 Install on Windows**..... 12
Install, first launch, a first-hour checklist, where the data lives and how updates work.
- 3 The window**..... 14
The sidebar, every page at a glance, scanning, tables and phones.

PART 1 • START HERE

1 What MediaLedger is

WHERE	WHO CAN OPEN IT	IT ANSWERS
Whole program	Everyone	What is this, what does it touch, and how does it work?

A ledger of everything on your media share, and a safe way to tidy it.

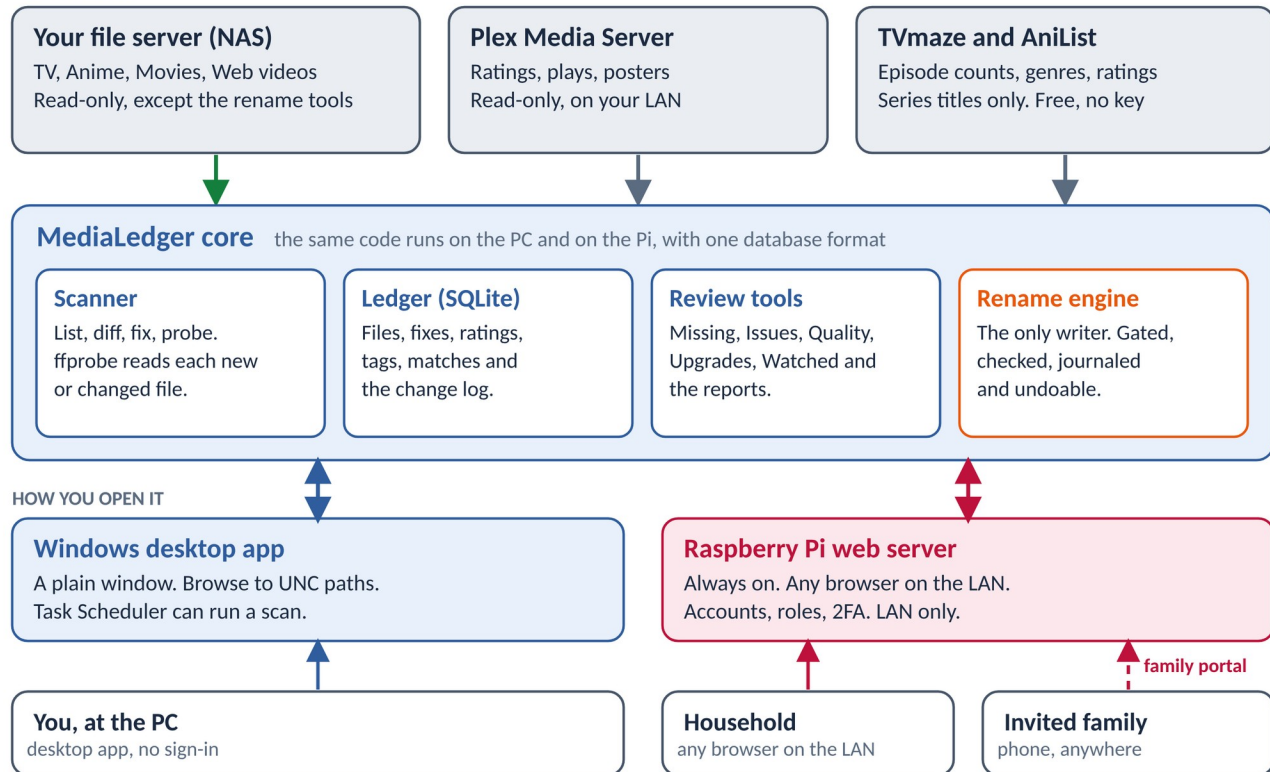
MediaLedger keeps a **ledger** of everything on your media share: every TV episode, anime episode, movie file and web video, what each file is technically made of, what changed since the last look, and what is missing. It reads the files directly over the network with the ffprobe tool, stores the results in a local database, and shows the picture in a set of pages. The same program runs two ways: as a Windows desktop app, and as a small always-on web server on a Raspberry Pi.

It is not a player, a downloader or a replacement for Plex. It sits beside Plex and answers the questions Plex is bad at: which series are missing episodes, which movies exist in two resolutions, which shows have no subtitles, what arrived on the NAS last week, and which files carry names that Plex will struggle with.

1.1 Three ways to use it

Windows desktop app A plain window on your PC. No sign-in: it runs as you. Browse to UNC paths such as \\nas\Media. Windows Task Scheduler can run a scan. Updates itself. Chapter chapter 2.	Raspberry Pi web server Always on. Any browser on the LAN. Accounts, roles, two-factor codes. LAN only by default. Plex webhook and Home Assistant status. One command to update. Chapter chapter 26.	Family portal A separate small site for people you invite. Browse, “what to watch tonight”, requests. Fixed fields only: no paths, no settings. Safe to publish on its own port. Chapter chapter 19.
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WHAT IT READS



Also: GitHub Releases (update check, ffmpeg download) · notifications by webhook or e-mail · Home Assistant status JSON

Figure 1. How the pieces fit. The same core runs on the PC and on the Pi, with one database format.

1.2 What it never does

- It never deletes a file. Files that disappear from the share are flagged **missing** in the ledger until you press a button to forget them.
- It never writes to the share unless you turn on one of the two rename tools, tick specific files, and confirm. Everything else is read-only.
- It never sends your library anywhere. The only network calls are: the update check against GitHub, the optional ffmpeg download, episode-count lookups to Tvmaze and AniList (series titles only), and your own Plex server on the LAN.
- The desktop app has no web server and no browser tab; it is a plain window. The optional Raspberry Pi build (chapter 26) is the same program served to your LAN behind a password.
- It never needs an online account, a subscription or an API key. The web server has its own local accounts (chapter 25); the desktop app has none.

Connection	Why	What leaves your network
GitHub Releases	The update check, and the optional ffmpeg download	Nothing about your library
Tvmaze and AniList	Episode counts, genres, ratings and posters	Series titles only. Free, no key
Your Plex server	Ratings, play counts, history and posters (read-only)	Nothing. It stays on your LAN

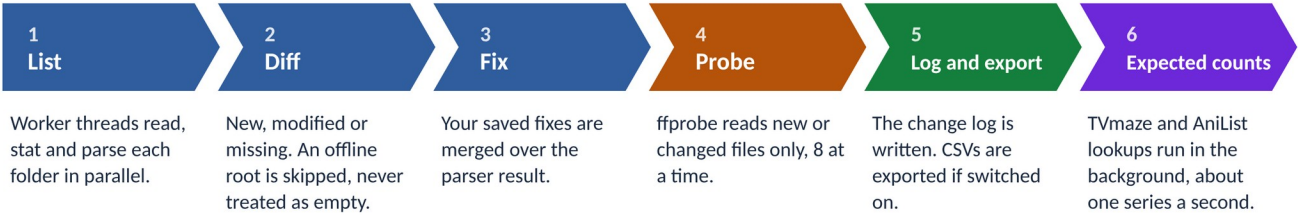
Connection	Why	What leaves your network
A webhook or mail server	Optional notifications you set up	The text of the message
Tailscale	Optional public address for the family portal	Portal traffic only

1.3 Core concepts

Term	Meaning
Root	A folder MediaLedger watches, with a type: TV, Anime, Movies, Web videos or Adult. Usually a UNC path such as <code>\\192.168.1.50\Media_Share\Anime</code> .
Scan	One pass over every enabled root: list files, compare with the ledger, parse names, probe new or changed files, write the change log, export CSVs.
Parse	Working out series, season, episode, title and year from the folder and file name. Fast and offline.
Probe	Running ffprobe on a file to read its real resolution, length, codecs, languages and subtitle tracks. This is the slow part; it only happens for new or changed files.
Fix	A manual correction you make to a parsed result. Fixes live in the database and are re-applied on every scan, so a corrected file stays corrected.
Change log	The record of what each scan found: added, removed, modified and returned files, plus errors.
Expected episodes	Per-season episode counts fetched from TVmaze (TV) or AniList (anime), used to say exactly which episodes you lack.
Placeholder	A literal word such as <code>Year</code> or <code>Source</code> that the movie naming engine writes into a name when it cannot prove the value. Deliberately visible so you can fill it in later.
Library type	TV, Anime, Movies or Web videos. A root's type decides which parser reads its names.
Title	One series or one movie: a row in the TV, Anime and Movies lists.
Card	A tile, chart or list on the Dashboard. You choose and arrange them.
Snapshot	A daily record of file count, free space and missing episodes that the trend cards draw.

1.4 How a scan works

A scan is one pass over every enabled root. The first one probes every file and takes a while; later ones only touch what changed. A scan starts when you press **Scan now**, on the schedule you set, when the folder watcher sees changes settle, or when Plex tells the server a new item arrived.



TIMINGS ON A GIGABIT SMB LINK TO A CONSUMER NAS, 24,628 FILES

25 min first full scan, everything probed 1 thread, 4 probes	2 min 40 s TV root alone, 11,419 files 4 threads, 8 probes	3 s rescan when nothing changed only stat calls, no probing
---	---	--

Figure 2. The six stages of a scan, and what each costs on a gigabit link.

NOTE A root that cannot be reached is logged as `root_offline` and skipped. Its files are never marked missing, so a NAS outage cannot be mistaken for a deletion.

PART 1 • START HERE

2 Install on Windows

WHERE	WHO CAN OPEN IT	IT ANSWERS
Windows PC	You	How do I install it and finish my first scan?

2.1 Install

1. Download `medialedger-<version>-setup.exe` from the Releases page of the repository.
2. Run it. It is a one-click, per-user install with no administrator prompt. The build is not code-signed, so Windows SmartScreen will show a blue box the first time: click **More info**, then **Run anyway**.
3. The app opens automatically when the installer finishes and is added to the Start menu.

2.2 First launch

On the first launch MediaLedger checks for `ffprobe`. If an `ffmpeg` install exists under `C:\ffmpeg`, via `winget` or `scoop`, or on your `PATH`, it is used. If nothing is found, the app downloads the latest static Windows build of `ffmpeg` from the BtbN project into its own data folder and points itself at it. A toast in the corner shows progress; it is about 100 MB.

Next, open **Settings → Library roots** and point each root at your shares. The defaults match the AxialForge NAS layout; change the paths if yours differ. Press **Save settings**, then **Scan now** in the bottom-left corner.

NOTE The first scan probes every file and takes a while: about 25 minutes for 24,000 files over gigabit SMB. Every later scan only probes new or changed files and finishes in seconds. You can close the window during a scan started by the scheduler, but a manual scan is cancelled if you quit.

2.3 Your first hour

1. Install and open the app (two minutes). Let it find or download `ffprobe`.
2. Settings → **Library roots**: point each root at its share and press **Save settings**.
3. Press **Scan now**. The first scan takes about 25 minutes for 24,000 files over gigabit.
4. Open the Dashboard (chapter 4). Check **Library health** and the counters in the sidebar.
5. Open **Issues** (chapter 10) and fix the few files the parser could not place. Open **Missing** (chapter 9) to see which episodes you lack.
6. Optional but wise: turn on the **nightly backup** to a folder on the NAS (section 21.4) and connect Plex (chapter 23).

2.4 Where your data lives

Item	Location (under <code>%APPDATA%\MediaLedger\</code>)
The ledger	<code>medialedger.db</code> : files, fixes, ratings, tags, matches, change log

Item	Location (under %APPDATA%\MediaLedger\)
Settings	settings.json
Application log	medialedger.log (the Log page shows its tail)
ffprobe, if downloaded	tools\ffprobe.exe
Database backups	medialedger.db.backups\ : a copy before every schema upgrade; the newest 10 are kept
CSV exports	exports\<timestamp>\ plus a latest\ copy
Posters	posters\, or the folder you chose in Settings
The program itself	%LOCALAPPDATA%\Programs\MediaLedger\

Because the data sits in your user profile and not in the program folder, reinstalling or updating never loses the ledger, your fixes or your ratings.

2.5 Updates

The installed app checks GitHub Releases on launch and every six hours, downloads a newer version silently and installs it the next time you close the app. The About page has a **Check for updates** button and shows the current state. No account or token is needed.

PART 1 • START HERE

3 The window

WHERE Left sidebar	WHO CAN OPEN IT Everyone; the pages you see depend on your role	IT ANSWERS Where is everything, and what do the badges mean?
------------------------------	---	--

The left sidebar is the only navigation. It is grouped into **Library** (what you have), **Review** (what needs attention), **Maintenance** (tools that change or export things) and **App**. Orange counters on Missing, Issues (problems, then duplicates), Requests and Movie names show how many items wait for you.

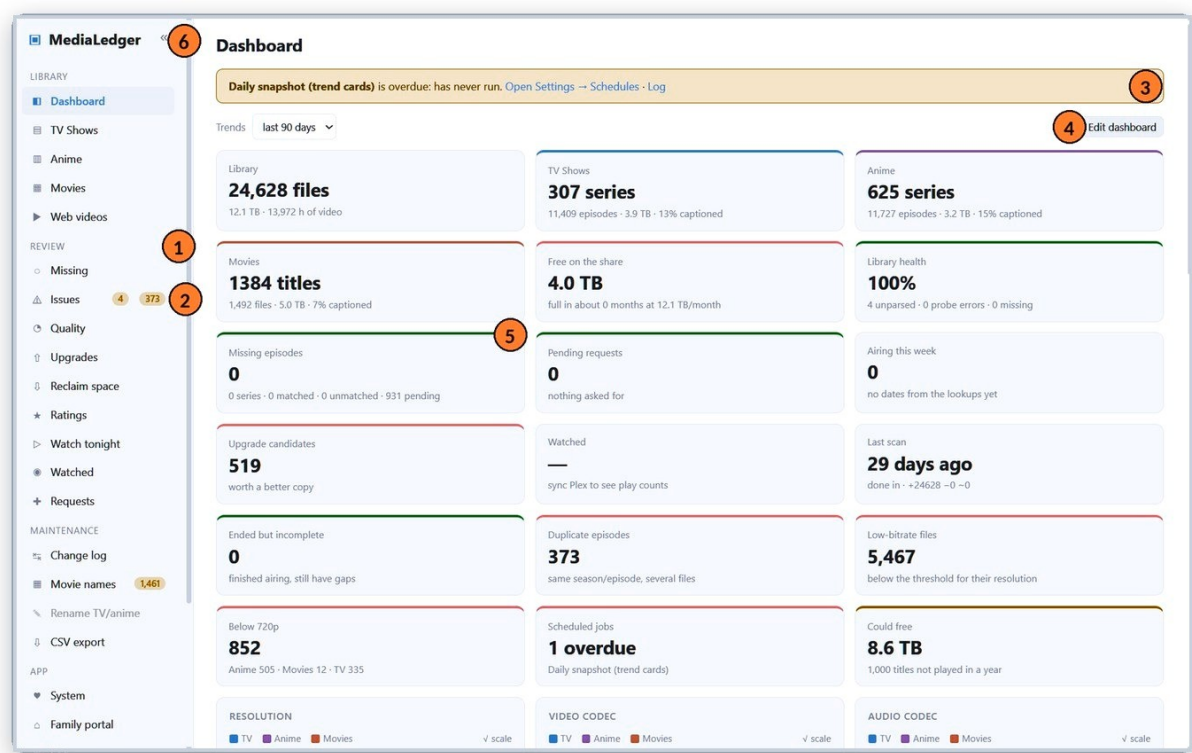
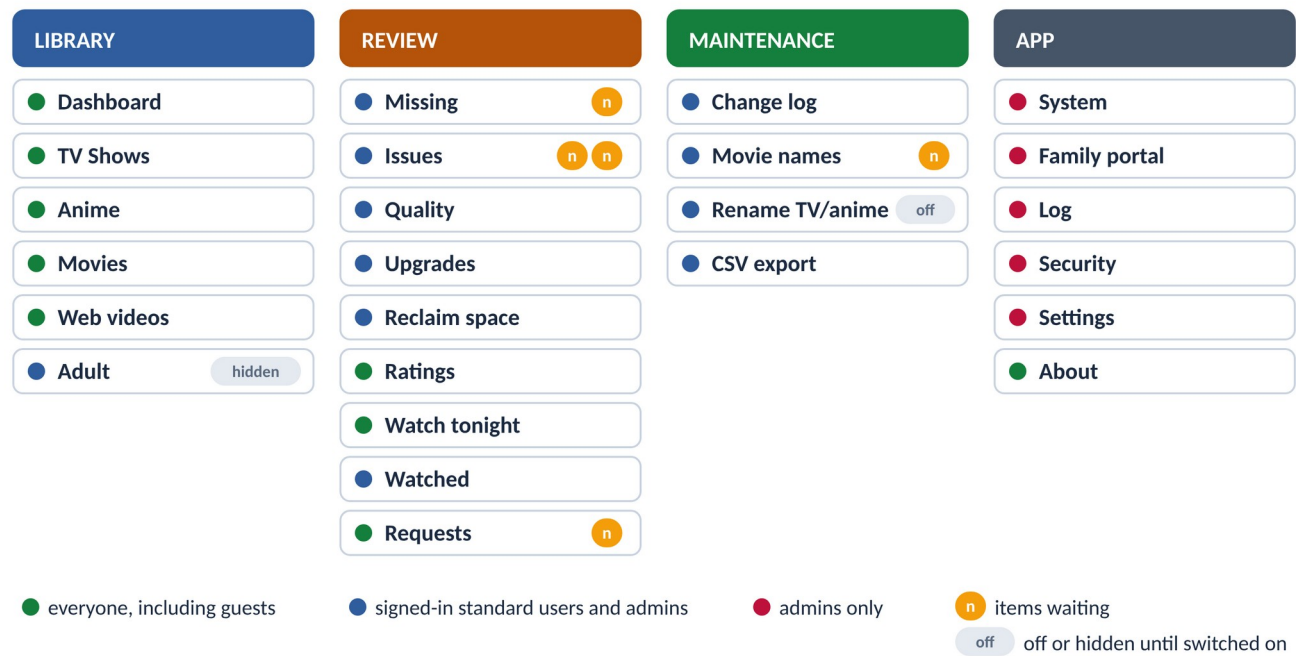


Figure 3. The Dashboard with its parts numbered. The table below explains each one.

Part	What it does
1	Sidebar groups Library, Review, Maintenance and App. The only navigation.
2	Counters Orange badges count what waits for you: Missing, Issues (problems, then duplicates), Movie names and Requests. A dot on System or Security means attention is needed.
3	Notices A banner above the cards, for example an overdue scheduled job or, on the Pi, a newer release.
4	Dashboard controls The trend period and Edit dashboard (section 4.7).
5	Colour rule The bar on a number tile is green, amber or red from thresholds you can change (section 4.5).

Part	What it does
6	« button

3.1 Every page at a glance



On the desktop app there are no accounts, so every page is open to you.

Figure 4. The sidebar, with who can open each page on the web server. On the desktop app every page is open to you.

Standard users can open the Review and Maintenance pages but only admins can run renames, fixes, exports and scans. Guests see library lists and statistics and can file requests. See [section 25.1](#) for the full table.

3.2 Scanning and the sidebar

- **Scan now** starts a manual scan. While a scan runs the button becomes **Cancel scan** and a progress bar shows the phase (listing, indexing, probing), the rate and an estimate of the time left.
- A second thin progress bar appears while expected-episode lookups or a Plex sync run in the background.
- The **Show adult content** switch appears only when an Adult root is configured. It is off on every launch.
- The version in the bottom-left corner shows "(dev)" when the app is run from source rather than installed.
- On the web server an account line at the bottom of the sidebar shows who is signed in, with **Sign in** / **Sign out**.

3.3 Tables, filters and sorting

Every table in the app can be sorted by clicking a column heading (click again to reverse) and filtered with the search box above it. Many rows are clickable: a series opens its episodes, a movie title opens its files, an episode or movie file reveals itself in Windows Explorer.

3.4 On a phone

- Below a certain width the sidebar becomes a slide-out menu: the ☰ button at the top opens it, and the top bar carries a **Scan** button.
- Dashboard cards widen by themselves, and the grid becomes one or two columns.
- The site ships a web-app manifest, so **Add to Home screen** opens it full screen like an app. Over plain HTTP with a self-signed certificate some browsers hide that option; see [section 25.5](#).

PART 2

Your library

Browse what you have: the dashboard, series, movies, web videos and the adult library.

IN THIS PART

- 4 Dashboard**..... 18
Tiles, charts, panels, colour rules, the storage forecast and how to edit the layout.
- 5 TV Shows and Anime**..... 22
The series list, episode pages, tags, sorting, the poster wall and where posters come from.
- 6 Movies**..... 26
One row per title, grouped by name and year, with every version side by side.
- 7 Web videos**..... 28
A root for downloaded web content, grouped by channel folder.
- 8 Adult library**..... 29
A root type that stays hidden everywhere until you switch it on for your own session.

PART 2 • YOUR LIBRARY

4 Dashboard

WHERE Sidebar → Library → Dashboard	WHO CAN OPEN IT Everyone. Some cards need a standard or admin account	IT ANSWERS How big is the library, is it healthy, and what changed?
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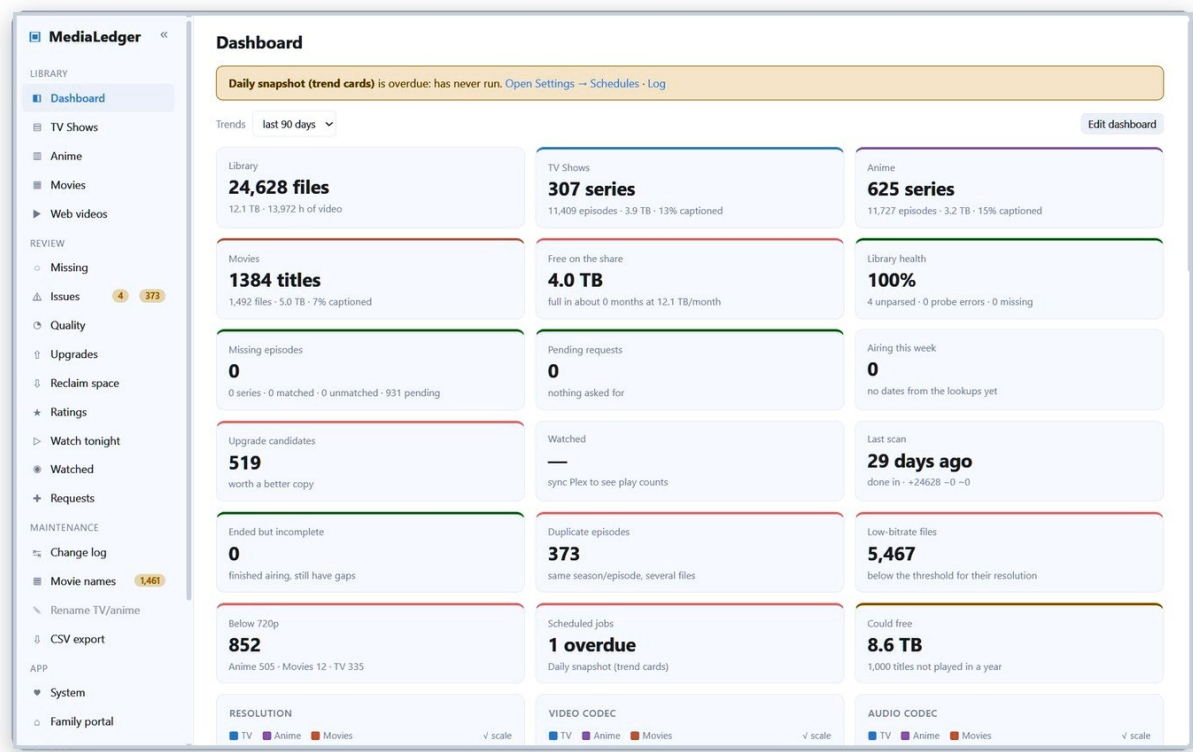


Figure 5. The Dashboard after a full scan.

The Dashboard is a summary of the whole ledger, computed live from the database. While nothing in the library changes the server answers from memory for up to a minute, so reopening the page is instant; it never serves cached numbers during a scan.

4.1 Top tiles

Tile	What it shows
Library	Total files, total size and total hours of video across every non-adult root.
TV Shows / Anime / Movies	Number of series or titles, number of files, size, and the share of files that have captions (embedded or sidecar subtitles).
Movie multiples	Titles that exist as more than one file, how many extra files that is, and their combined size.
Library health	100% minus the share of files that are unparsed, failed to probe, or are missing.

Tile	What it shows
Captions	Files with any subtitle track or sidcar.
Below 720p	Files whose probed resolution is 576p, 480p or SD.
Avg bitrate	Average video bitrate, overall and per library.
Last scan	When it ran, how long it took, and its added / removed / modified counts.
Manual fixes	How many corrections you have saved.
Last export	When CSVs were last written and how many files.
Missing episodes	Episodes you lack according to TVmaze / AniList, how many series have gaps, and how many series are matched, unmatched or still pending.
Duplicate episodes	Season/episode combinations that exist as several files.
Mixed-quality series	Series with more than one resolution.
Low-bitrate files	Files under the bitrate threshold for their resolution (thresholds are in Settings).
Undefined audio language	Files whose audio track carries no language tag.
Folder watch	Whether the optional folder watcher is on and when it last saw a change.

4.2 Charts

Eight stacked bar charts break the probed files down by resolution, video codec, audio codec, container, audio language, subtitle language, frame rate and dynamic range. Each bar is split by library (blue TV, purple Anime, orange Movies); hover a segment for the exact count.

4.3 Panels

- **Recently added** – the newest files by first-seen time, with a link to the change log.
- **Largest series** and **Largest movie files** – by size on disk.
- **Most missing episodes** – the series with the biggest gaps, with the exact episodes listed; whole missing seasons collapse into ranges such as "S6–S12 entirely".
- **Scan history** – the last eight scans with status, trigger, thread count, duration and counts.

4.4 What tags and lookups add

A third row of tiles covers what arrived with tags and lookups: **Pending requests**, **Airing this week** with the next episode named, **Watched** (Plex play counts across linked files, a click away from Watch tonight), **Genres known**, **Your tags**, **Upgrade candidates** and **Ended but incomplete**. Below the codec charts, three panels chart **Genres** by library, **Anime sub / dub**, and **Your tags**. The TV, Anime and Movies lists open with the same idea: Watched, Complete, Sub / dub, Top genres and Your tags tiles next to the size and runtime.

4.5 Colour rules and drill-down

Bar charts use a **square-root scale**, so a codec with 25,000 files no longer flattens the ones with 40; the exact count and the share of the total sit beside every bar, and hovering a bar shows the per-library split. Clicking a bar segment opens that library's list already filtered to the value (`#anime?q=720p`). Donuts show shares with the value in the centre, and the trend cards draw a line from the **daily snapshot** the server takes after each finished scan (or at 03:05): files, free space and missing episodes over time.

Number tiles carry a **colour rule**: green, amber or red from thresholds. Every rule ships with a default (missing episodes turn amber at one and red at a hundred, library health turns amber under 97 %). Hover a tile and press the  to change it. The dialog has an **editor level** for your account: Simple hides every knob, Standard keeps the shipped defaults, Advanced lets you set the thresholds and the direction. Preferences are stored per account on the web server, and once for the desktop app.

Tile	Direction	Amber at	Red at
Library health	higher is better	below 97 %	below 90 %
Missing episodes	lower is better	1	100
Pending requests	lower is better	1	5
Upgrade candidates	lower is better	1	20
Ended but incomplete	lower is better	1	10
Duplicate episodes	lower is better	1	50
Low-bitrate files	lower is better	1	1,000
Below 720p	lower is better	1	500
Free on the share (months of space left)	higher is better	below 12	below 3

4.6 Storage forecast

The **Free on the share** tile and the Storage panel at the bottom use the date each file was first seen to work out how much the library grows per month, average the last three complete months, and divide the free space on your roots by it. The tile turns amber under a year and red under three months. It is an estimate from your own history, so a one-off bulk import skews it for a quarter.

4.7 Editing the dashboard

Every element of the Dashboard is a **card** from a catalog, laid out on a grid six columns wide. Press **Edit dashboard** and each card gets a handle:

- **Drag** a card to move it, or use ▲ ▼ on a phone.
- **S, M, L, XL** set the width: one, two, three or all six columns.
-  opens the card's options. ✕ removes the card.
- **Add card** opens the catalog. A card can be added more than once, for example the same trend over a month and over a year.
- **Reset to default** restores the shipped layout. **Done** saves.

The layout is saved **per account** on the web server, so your phone and your PC show the same arrangement and each person has their own; guests see the admin's. The desktop app keeps one layout. On narrow screens cards widen by themselves and the grid becomes one or two columns.

A card's options depend on the **options shown** level, chosen at the bottom of the dialog and remembered for your account. **Simple** offers the title and the size. **Standard** adds the period for trend cards and the number of rows for lists. **Advanced** adds the colour thresholds of number tiles. Thresholds you set before version 2.0 with the gear on a tile carry over. The trend period picker above the grid sets the period for every trend card that does not have its own.

NOTE The editor draws from the data it already holds while you edit and loads fresh when you press **Done**. On a Raspberry Pi that makes editing smooth; it used to recompute the whole dashboard on every move.

4.8 The card catalog

The catalog has 53 cards in eight groups. Cards marked **standard** or **admin** are hidden from people with a lower role; the rest are visible to guests. A new account starts with 33 of them in the default layout.

Group	Cards
Library (7)	Library · TV Shows · Anime · Movies · Movie multiples · Genres known · Your tags
Health (6)	Library health · Missing episodes · Ended but incomplete · Duplicate episodes · Manual fixes · Overdue jobs (admin)
Quality (7)	Below 720p · Low-bitrate files · Upgrade candidates · Mixed-quality series · Captions · Average bitrate · Undefined audio language
Activity (7)	Last scan · Last export · Folder watch · Pending requests · Airing this week · Scan history · Recently added
Storage (5)	Free on the share · Storage added per month · Reclaimable space (standard) · Largest series · Largest movie files
Watching (7)	Watched · Watched (chart) · Plays per day (standard) · Plays by person (standard) · Next up (standard) · Anime sub / dub · Most missing episodes
Charts (10)	Resolution · Video codec · Audio codec · Container · Audio languages · Subtitle languages · Frame rate · Dynamic range · Genres · Your tags
Trends (4)	Files over time · Free space over time · Missing episodes over time · Pending requests over time

PART 2 • YOUR LIBRARY

5 TV Shows and Anime

WHERE Sidebar → Library → TV Shows or Anime	WHO CAN OPEN IT Everyone. Only admins can change fixes and matches	IT ANSWERS What series do I have, in what quality, and how complete are they?
---	--	---

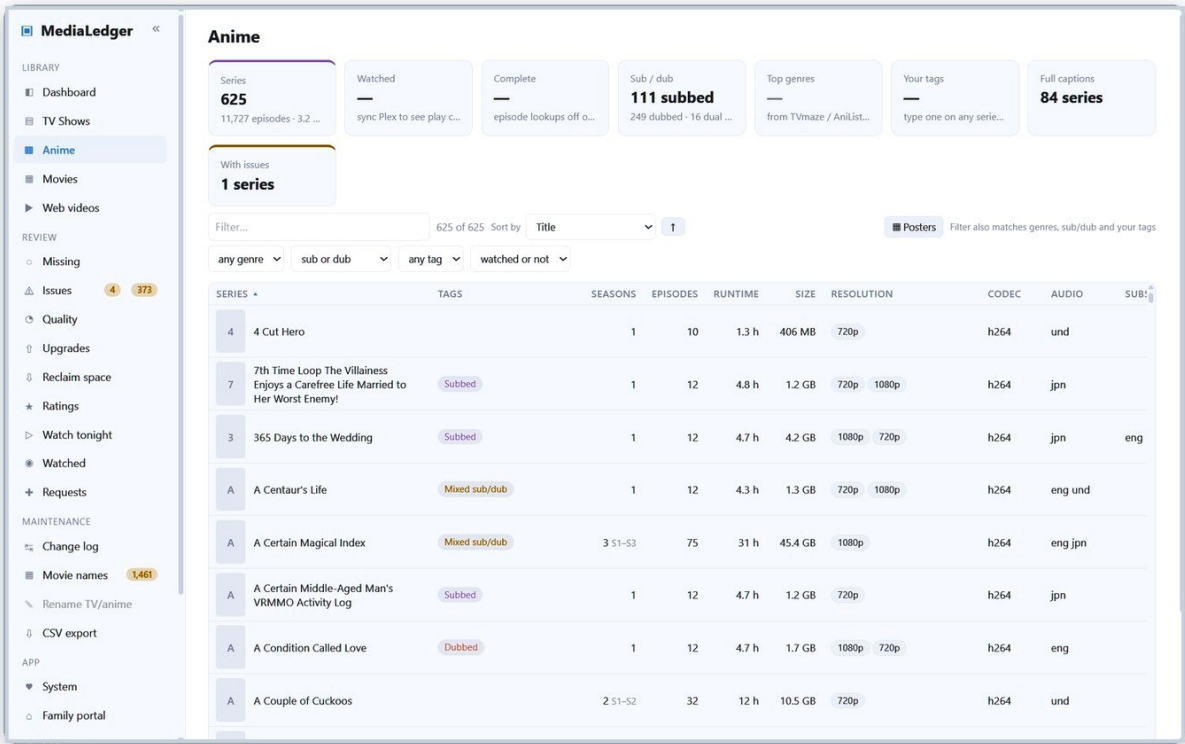


Figure 6. The Anime list. TV Shows looks the same.

One row per series. The summary strip counts series, episodes, size, runtime, how many series have full captions and how many have parse issues.

Column	Meaning
Seasons	Distinct season numbers on disk, with the range when it is not 1..N.
Episodes	Files that belong to the series.
Runtime / Size	Totals from the probe.
Resolution / Codec	Every resolution and codec present.
Audio / Subs	Union of audio and subtitle languages across the series.
Rating	The online average (TVmaze or AniList) once the series has been matched.
Mine	Your own 0–5 stars. Click a star to set it; click the same star again to clear. A small "P4.5" beside it is your Plex rating when Plex is synced.

Column	Meaning
Watched	Share of episodes Plex has marked played (needs a Plex sync).
Captions	Share of episodes with subtitles: green 100%, amber partial, red none.
Missing	Episodes you lack out of the expected total, or "complete", or "no match" when the online lookup could not find the series.
Issues	Unparsed or unprobed files in the series.

5.1 Episode detail

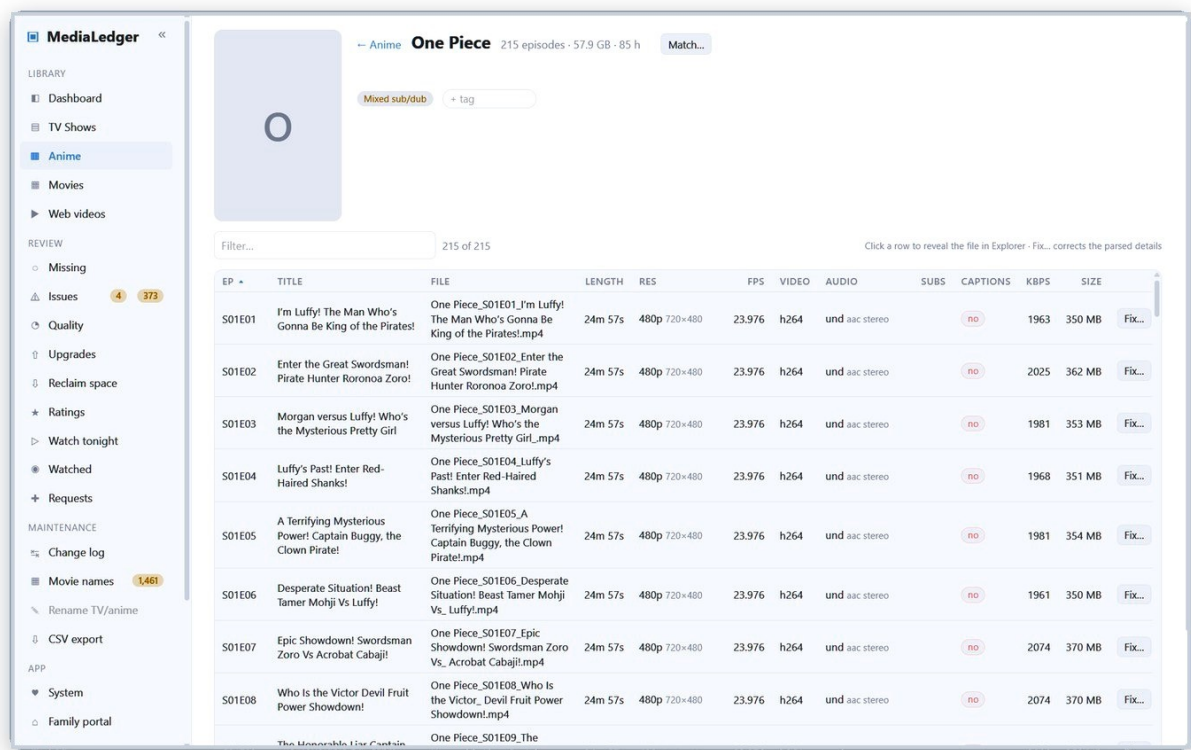


Figure 7. A series opened from the Anime list, with the expected-episode grid at the top.

Click a series to open it. The header shows the totals and a **Match...** button (see [section 9.3](#)). When expected counts exist, a grid shows every season with each episode number as a green (present) or red (missing) cell.

The table lists every file with its parsed episode number, title, file name, probed length, resolution and pixel size, fps, video codec with bit depth and HDR badge, audio languages and codecs, subtitle tracks, captions flag, bitrate and size. Badges mark files that are **missing** from disk or have a manual **fixed** override. Click a row to reveal the file in Explorer; click **Fix...** to correct its details ([section 10.2](#)).

5.2 Tags: genres, sub/dub and your own

Every series and movie carries three kinds of tag, shown in a **Tags** column on the lists and as a strip under the title on its own page. The filter box matches them, so typing **comedy**, **dub** or **kids** narrows the list, and the dropdown row under it filters by one genre, sub/dub state, tag, or watched state (from Plex) at a time.

Kind	Where it comes from	Examples
Genres (grey)	The same TVmaze / AniList lookup that fetches episode counts also returns genres, and AniList adds its crowd-ranked tags. Fetched automatically with the next lookup; series matched before this version are filled in once in the background. Movies take their genres from Plex when synced.	Action, Slice of Life, Isekai, Documentary
Sub / dub (coloured)	Worked out on this machine from the audio and subtitle languages fprobe recorded: Japanese audio with English subtitles is Subbed , English audio on anime is Dubbed , both audio tracks is Dual audio , and a series whose episodes disagree is Mixed . Nothing is fetched.	Subbed, Dubbed, Dual audio
Your tags (blue)	Anything you type into the + tag box on a title page, Enter to add, × to remove. Earlier tags are suggested as you type. Stored with the title like your star rating, so they survive scans and renames.	kids, Christmas, watch with Sarah, rewatch

All three are exported in the series and movie title CSVs (genres, audio_type, tags). Guests see tags but cannot edit them; standard users and admins can.

5.3 Sorting and the poster wall

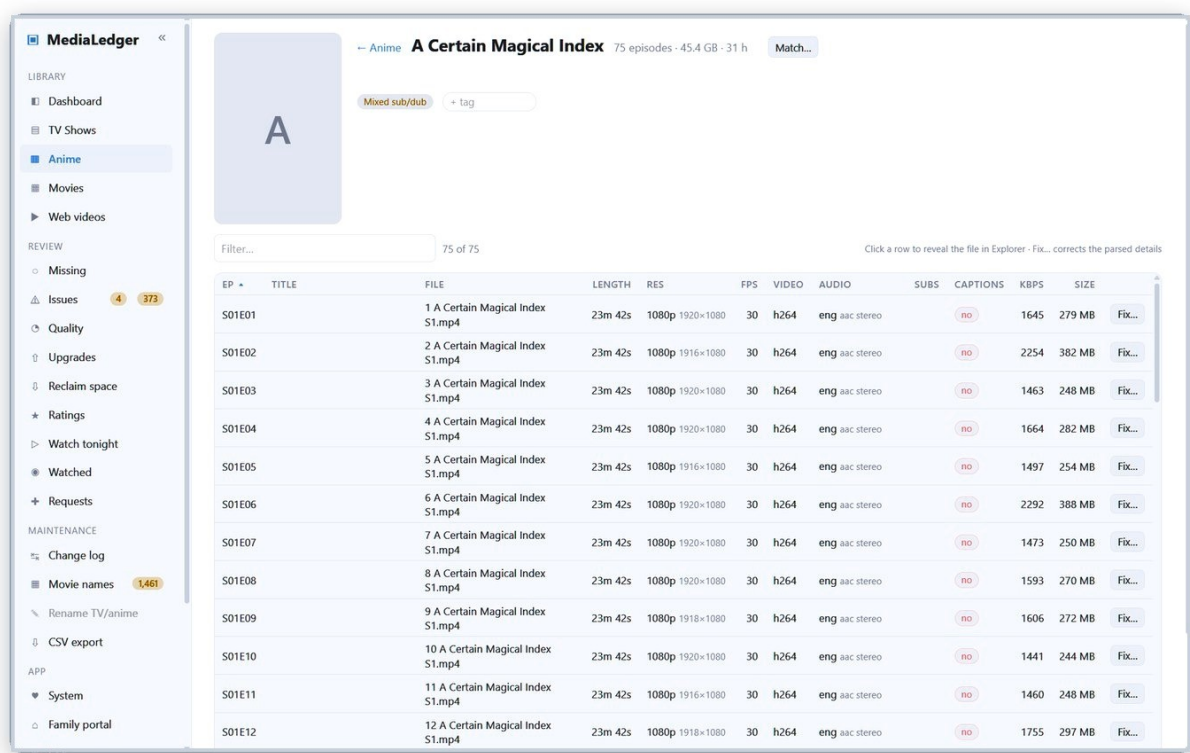


Figure 8. A series page with its poster. The lists carry a thumbnail and can switch to a poster wall.

Sorting a list. Beside the filter box, **Sort by** lists what the TV, Anime and Movies pages can be ordered by: title, year, the ratings (online, Plex audience and critic, your own), unwatched episodes, last episode added, date added, date viewed, plays, episodes, seasons, duration, size, resolution, bitrate, missing episodes, and **Randomly**; Movies adds progress and versions. The arrow button next to it flips between ascending and descending. Clicking a column

heading still sorts too. The choice is remembered per list on the device you set it on. "Date added" is the earlier of when MediaLedger first saw the file and the file's own date, so an old library does not all look new.

On the poster wall each tile shows the seasons and episodes on hand (for a movie: length and quality) under the title. Rest the pointer on a tile for a card with the rest: runtime and size, quality, missing episodes, how much is watched and when last, ratings, the next episode date, when it was added, genres and your tags.

Every series and movie can carry a **poster**. They appear as a thumbnail beside the title in the TV, Anime and Movies lists and in Watch tonight, large on the title page, and throughout the family portal.  **Posters** in the toolbar of a list turns the table into a wall of posters; search, the filters and sorting keep working, and  **Table** switches back. A title without art shows its first letter instead.

5.4 Where posters come from

Posters come from **Plex** first, asked for at the size needed so each file is small, and from **AniList** or **TVmaze** for series that Plex does not have but the episode lookup matched. New titles are fetched after each scan and each Plex sync; Settings → **Posters** has **Fetch missing posters**, **Fetch all again** and a count by source. A title with no art anywhere is asked for again after two weeks. **Adult titles never get a poster**.

Where they are kept. By default next to the database. Set the **posters folder** to a folder on the NAS to share one set between the desktop app and the web server; a library of 2,300 titles takes about 100 MB. On the Raspberry Pi the folder must be a mounted path such as `/mnt/medialedger/Share/MediaLedger/Posters`, writable by the `medialedger` user. Files are named by a fingerprint of the title, so nothing in a title can produce an odd file name. Removing a title from the library removes its poster on the next run.

PART 2 • YOUR LIBRARY

6 Movies

WHERE Sidebar → Library → Movies	WHO CAN OPEN IT Everyone	IT ANSWERS Which movies do I have, and which exist in more than one version?
--	------------------------------------	--

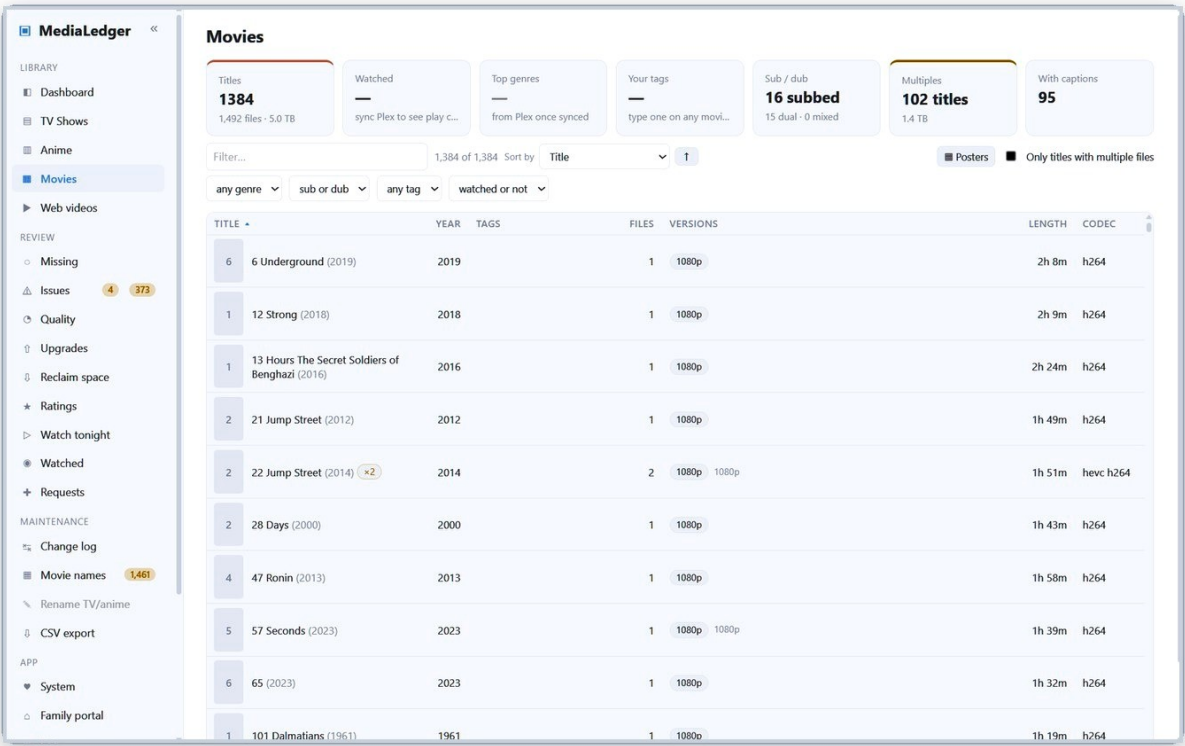


Figure 9. The Movies list. A x2 badge marks titles with two files.

One row per title. Files are grouped by a normalised title plus year, so `Pacific Rim (2013).mp4` and `Pacific Rim (2013) [4k].mkv` become one title with two versions. The **Only titles with multiple files** checkbox narrows the list to those.



Figure 10. The files behind one title, side by side.

Click a title to see its files side by side: edition tag, length, resolution, fps, video codec and profile, audio, subtitles, captions, bitrate, size and container. Each row has **Fix...** for corrections and reveals the file in Explorer when clicked.

TIP A movie file whose name carries a season or episode marker, such as `S01E02` or `1x02`, is listed under **Episodes** filed under **Movies** on the Issues page (section 10.4).

PART 2 • YOUR LIBRARY

7 Web videos

WHERE Sidebar → Library → Web videos	WHO CAN OPEN IT Everyone	IT ANSWERS What have I downloaded from the web, by channel?
--	------------------------------------	---

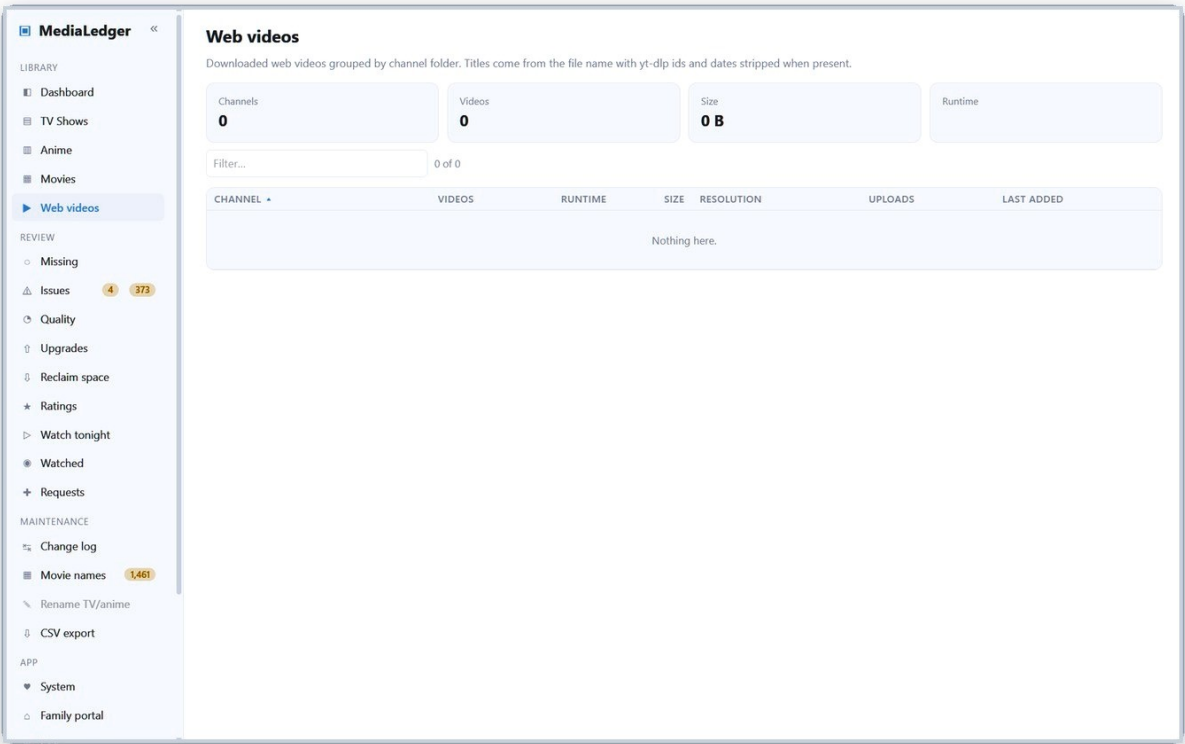


Figure 11. Web videos grouped by channel folder.

A root of type **Web videos** is for downloaded web content, typically from yt-dlp. The top-level folder is treated as the channel; loose files at the root fall under "(no channel)". The file name becomes the title after stripping a trailing `[videoId]` and any `YYYYMMDD` date. When an id is present the video list links to it on YouTube.

The channel list shows video count, runtime, size, resolutions, the upload-date range and when the last file arrived. Open a channel to see its videos with every probed field. A channel can be rated with stars from its header.

PART 2 • YOUR LIBRARY

8 Adult library

WHERE	WHO CAN OPEN IT	IT ANSWERS
Sidebar → Library → Adult (hidden until switched on)	Admins and standard users; guests never	How do I keep adult content apart from the rest?

A root of type **Adult** holds content you want kept apart. MediaLedger classifies each file under it as anime, TV or movie from its folder and name, so the ordinary series and movie views and tools all work on it. What differs is visibility:

- Adult files are excluded from every list, count, chart and query until the **Show adult content** switch at the bottom of the sidebar is on.
- The switch is **off on every launch**. Nothing remembers it.
- While it is on, an **Adult** tab appears under Library with its own summary, resolution chart, recently added, series list, movie list and unparsed list. Series open in the normal episode view.
- Adult files are left out of CSV exports unless **Settings → Adult content → Include in CSV exports** is ticked.
- Screenshots and the README never include adult content.

Under Settings → Adult content you can also choose whether files that match none of the classification patterns are treated as anime or TV.

NOTE On the web server the switch is **per session**: one person switching it on does not show it to anyone else, and guests and the family portal never see adult titles. Posters are never fetched for adult titles.

PART 3

Review

What is missing, wrong, low quality or never watched, and how to fix it.

IN THIS PART

9 Missing episodes	31
Expected counts, airing dates, the Match dialog and per-series collecting choices.	
10 Issues	33
Problems, the Fix dialog, duplicate decisions and episodes stored in the movie folders.	
11 Quality and upgrades	36
The quality report, bitrate thresholds, and how the upgrade score ranks titles.	
12 Ratings and Watch tonight	39
Your stars beside the online and Plex scores, and a one-click “pick for me”.	
13 Watched and reclaim space	41
Plex play history for everyone, the next episode for each viewer, and big titles nobody plays.	

PART 3 • REVIEW

9 Missing episodes

WHERE Sidebar → Review → Missing	WHO CAN OPEN IT Standard users and admins. Match and Collect are admin actions	IT ANSWERS Which episodes do I lack, and what airs next?
--	---	--

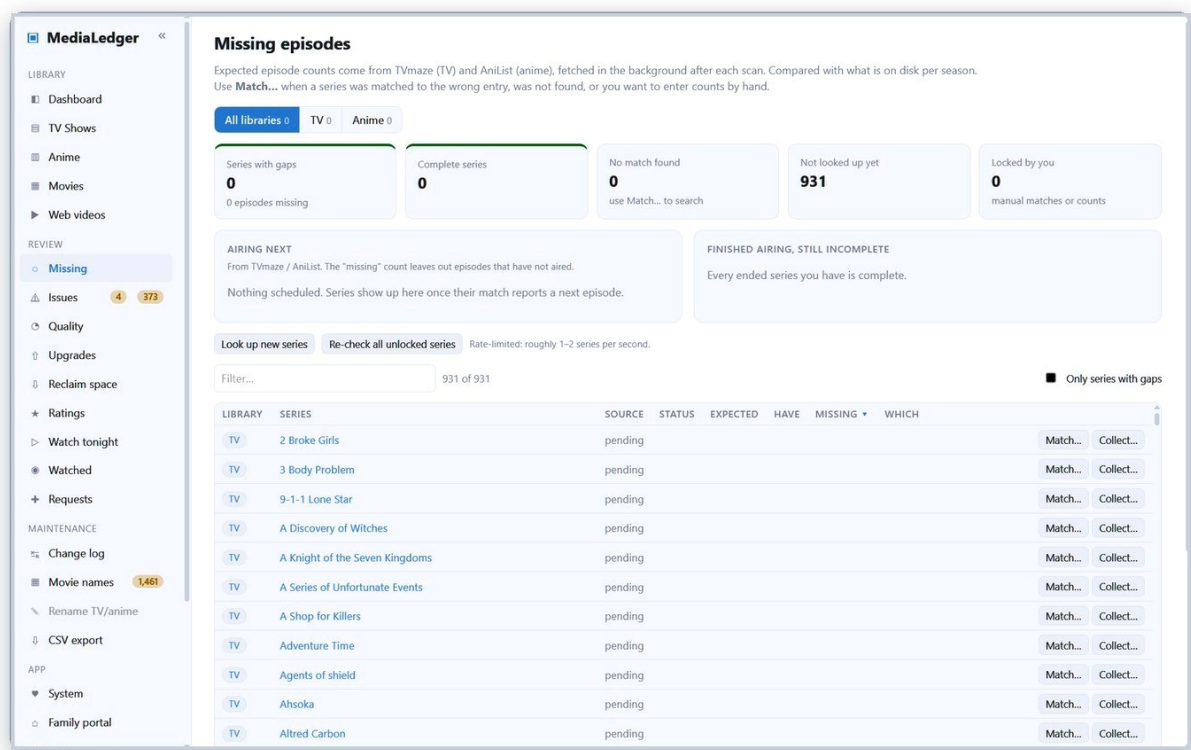


Figure 12. Series ranked by how many episodes are missing.

One library at a time. The buttons at the top of the page, **All libraries**, **TV** and **Anime**, narrow the whole page to one library: the tiles, the two airing cards and the table. Each button shows how many of its series have gaps. **Only series with gaps** hides the complete and unmatched series. Both choices are remembered on the device.

9.1 How the counts work

After each scan MediaLedger looks up every series it has not seen before on TVmaze (TV) or AniList (anime) and stores the number of episodes in each season. Both services are free and need no key; lookups run at about one series per second in the background, and airing series are re-checked every two weeks. Comparing those counts with what is on disk, per season, gives an exact list of what you lack.

Column	Meaning
Source	tvmaze, anilist, manual or none. A lock badge means you chose the match or entered counts yourself; automatic re-checks never overwrite a locked series.

Column	Meaning
Status	Running / Ended (TVmaze) or RELEASING / FINISHED (AniList).
Expected / Have / Missing	Regular-season episodes only; specials (season 0) are not counted.
Which	The missing episodes per season. Whole missing seasons collapse into ranges.
absolute numbering	A badge shown when a season on disk is numbered far beyond its expected length (for example One Piece folders numbered 62–77). That season is skipped rather than reported as missing.

9.2 Airing next

Two panels above the table use the same online match. **Airing next** lists every series whose match reports an upcoming episode, soonest first, with the date and the episode label and whether your copy is otherwise up to date; the next seven days are highlighted. Episodes that have not aired yet are never counted as missing. **Finished airing, still incomplete** lists series the source marks as ended while you still lack episodes: those gaps will not fill themselves. Both feed the daily summary ([chapter 20](#)) and the Home Assistant status.

9.3 The Match dialog

Press **Match...** on any series (here, on the series page, or in the lists) when a series was matched to the wrong entry, was not found, or you know the counts yourself. The dialog shows the current match, lets you search either TVmaze or AniList regardless of library type, and lists candidates with year, format and episode count; click one to use it. Below that you can type counts per season by hand, declare that the series has no expected counts, or return a locked series to automatic.

NOTE AniList numbers anime by cour, so a split season shows up as separate "Part 2" entries. MediaLedger merges those into the same season automatically. If a series still lines up better with broadcast seasons, switch its source to TVmaze in the Match dialog.

The two buttons above the list start a background pass: **Look up new series** for anything not yet looked up, **Re-check all unlocked series** to refresh everything that you have not locked.

9.4 What you are collecting

A long runner you joined late can swamp the Missing page: a thousand early episodes you never meant to keep. **Collect...** on a series row sets what counts as missing for that one series. **Everything** is the default. **From season S episode E onward** drops everything earlier; shows numbered straight through use season 1 and the episode number. **Mute this series** counts nothing. The row keeps a badge (from `S1E1100, muted`) whose tooltip gives the full count, and the dashboard, the sidebar count, the daily summary and the Home Assistant status all follow the policy. No file is touched.

PART 3 • REVIEW

10 Issues

WHERE Sidebar → Review → Issues	WHO CAN OPEN IT Standard users and admins can look. Only admins fix	IT ANSWERS What could the scan not settle, and which files are duplicated?
---	---	--

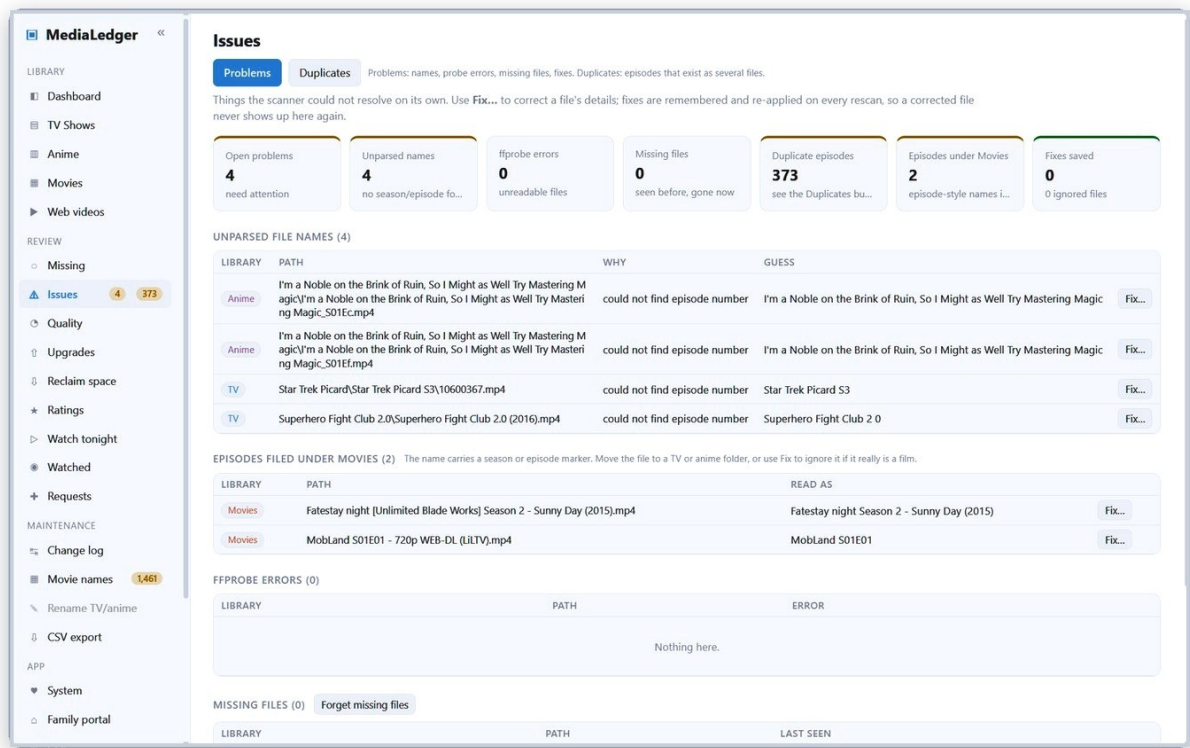


Figure 13. The Issues page with Problems selected. The Duplicates button switches to the second view.

Issues gathers everything a scan could not settle by itself behind two buttons at the top of the page: **Problems** (file names the parser could not place, ffprobe errors, missing files and your saved fixes) and **Duplicates** (episodes that exist as more than one file). The sidebar shows two counters, problems first and duplicates second. The old `#problems` and `#duplicates` links still open the matching view.

10.1 Problems

The summary strip counts open problems. Sections below list unparsed file names with the parser's best guess, ffprobe errors, missing files, and your saved manual fixes (including ignored files). The duplicate tile only counts; the list itself is under the Duplicates button.

- **Forget missing files** removes the records of files that have vanished. Until you press it they stay flagged, which protects you from a NAS outage being mistaken for a deletion.
- A root that cannot be reached during a scan is logged as `root_offline` and skipped entirely; its files are never marked missing.

NOTE Forgetting missing files asks for your password again on the web server, and the action is written to the security log.

10.2 The Fix dialog

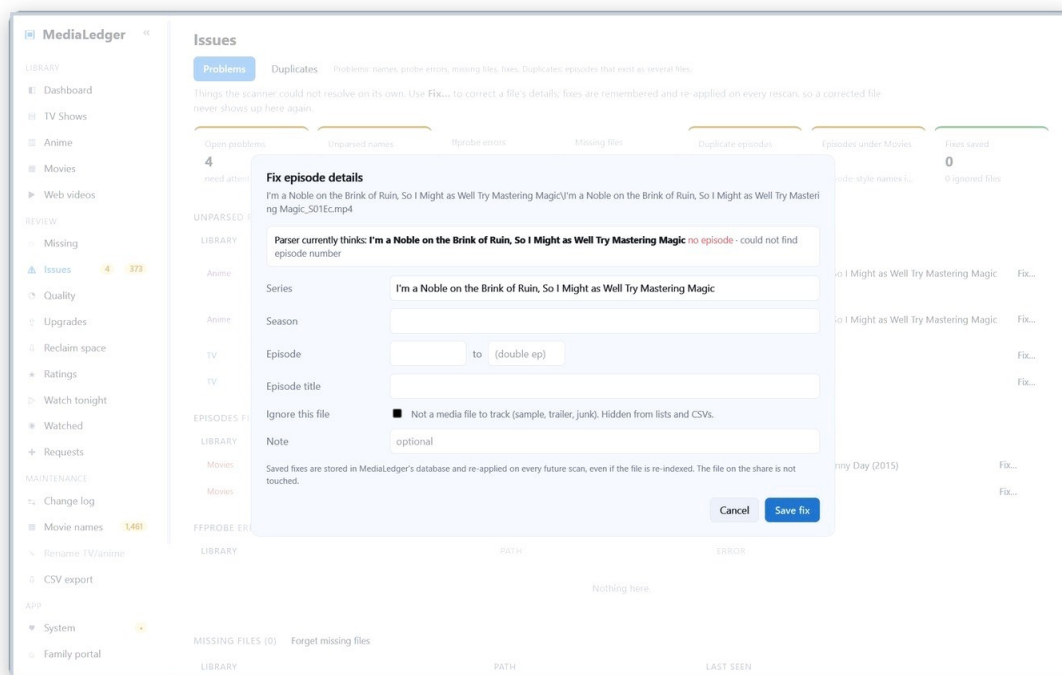


Figure 14. Correcting an episode that the parser could not place.

Fix... appears on every file row in the app. The dialog states what the parser currently thinks, then lets you set the correct values:

- Episodes: series (with suggestions), season, episode and an optional end episode for double episodes, episode title.
- Movies: title, year, edition tag, and **Source** (Web or Rip) for the naming engine.
- **Ignore this file** hides a file from every list, CSV and rename plan without touching it on disk.
- A free-text note.

Saved fixes are stored by root and relative path, re-applied on every future scan, and follow the file through the app's own rename tools. **Remove fix** returns a file to what the parser says.

10.3 Duplicates

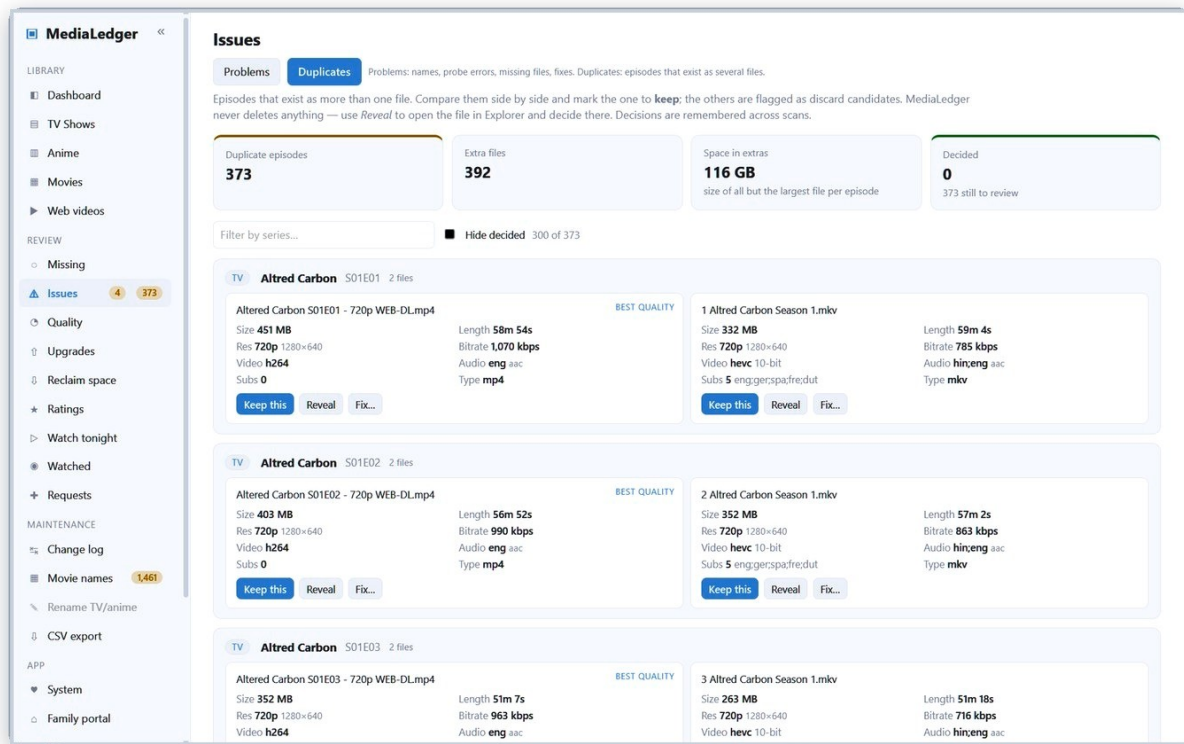


Figure 15. Two files for the same episode, side by side.

Every season/episode that exists as more than one file is shown as a group, each file as a card with size, length, resolution, bitrate, codec, HDR, audio, subtitles and container. The card with the highest resolution and bitrate is marked **best quality**.

Press **Keep this** on the file you want. It is marked keep, the others are marked discard candidates and drawn faded, and the group moves to the decided state. Nothing is deleted: use **Reveal** to open a file in Explorer and delete it there if you wish. Decisions are stored with the file and survive scans and renames; **Clear decision** undoes them. **Hide decided** narrows the list to groups still waiting.

10.4 Episodes filed under Movies

Episodes filed under Movies lists files in a movie folder whose name carries a season or episode marker (**S01E02**, **1x02**, **Season 2**). They show up as odd movies everywhere else, so move them to a TV or anime folder; if one really is a film, use **Fix** to ignore it. A bare "Episode 1" is not flagged, so the Star Wars films are safe.

PART 3 • REVIEW

11 Quality and upgrades

WHERE Sidebar → Review → Quality, and Upgrades	WHO CAN OPEN IT Standard users and admins	IT ANSWERS Which files look technically off, and which titles deserve a better copy?
--	---	--

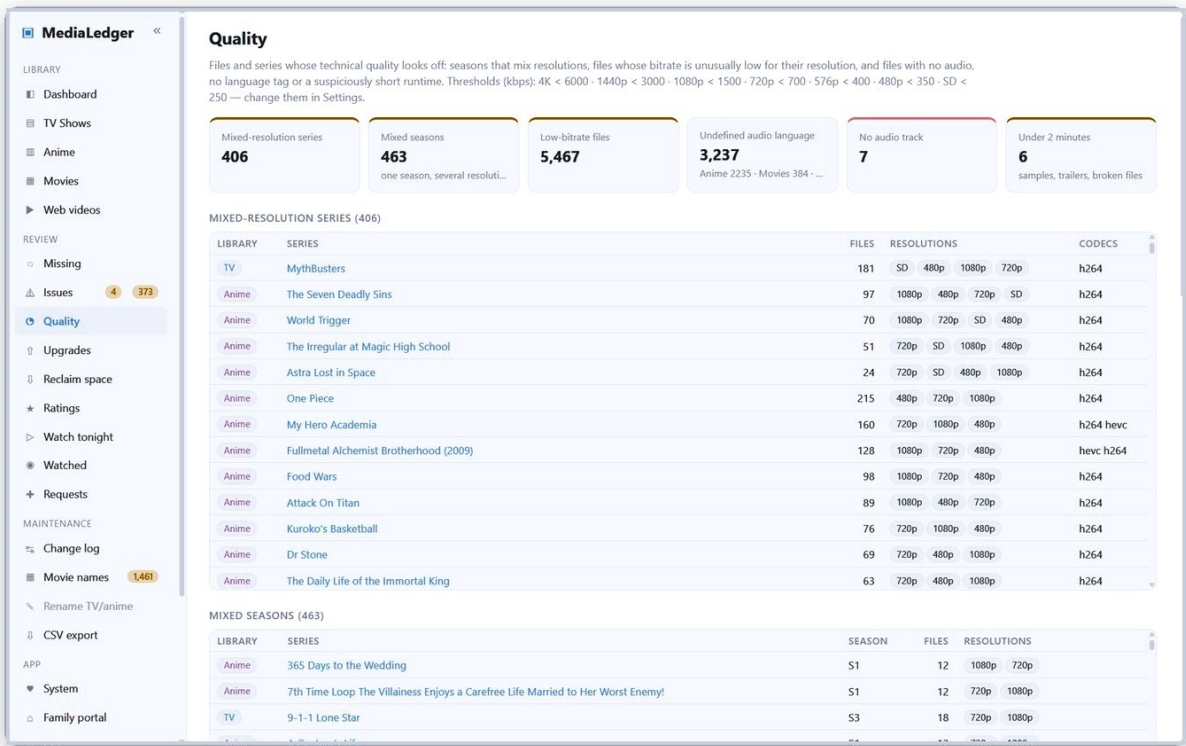


Figure 16. The Quality report.

Lists files and series whose technical quality looks off:

- **Mixed-resolution series** and **Mixed seasons**: a series, or a single season, that holds more than one resolution.
- **Low-bitrate files**: files whose video bitrate is below the threshold for their resolution (table below).
- **No audio track**, **Undefined audio language** and **Under 2 minutes**: usually samples, trailers or damaged files. Use **Fix...** to ignore a file that is not real media.

Resolution	4K	1440p	1080p	720p	576p	480p	SD
Default minimum (kbps)	6000	3000	1500	700	400	350	250

Change the numbers under Settings → Quality thresholds. They also drive the **Low-bitrate files** card and the upgrade score.

11.1 Upgrade candidates

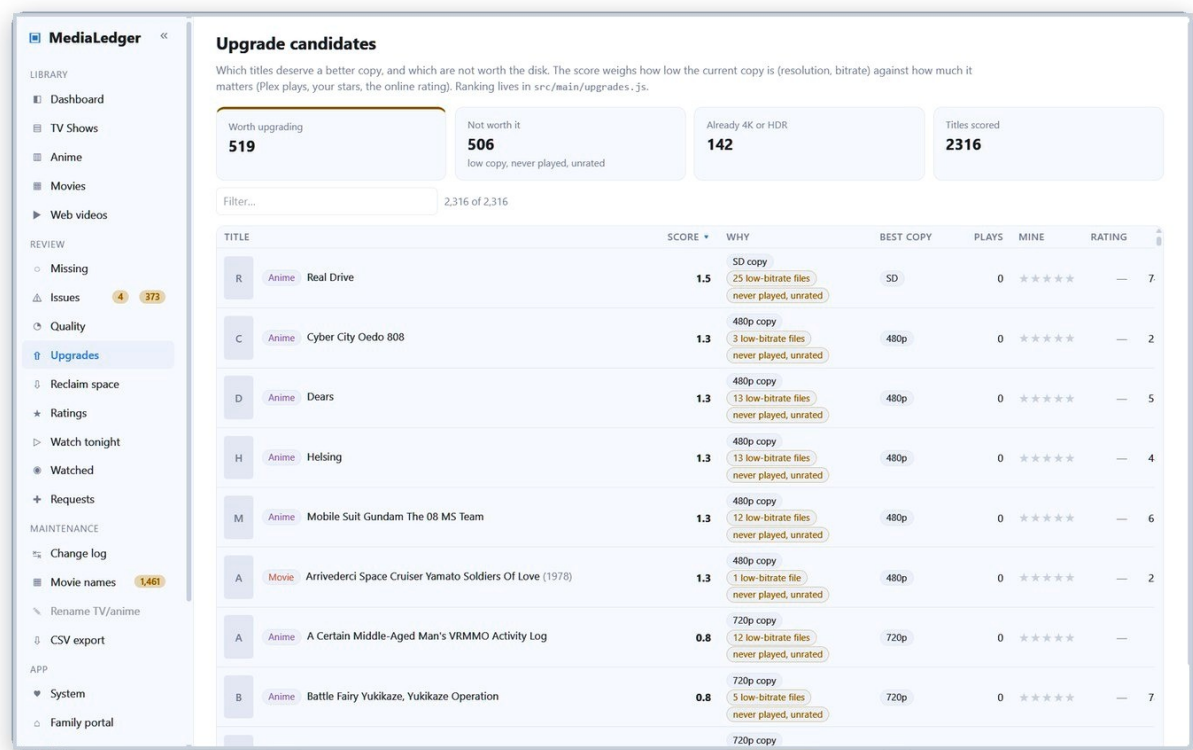


Figure 17. The Upgrades page: titles ranked by how much they deserve a better copy.

The **Upgrades** page ranks every title by how much it deserves a better copy: how low the current copy is (resolution, low-bitrate files) against how much it matters (Plex plays, your stars, the online rating). Each row shows the score and plain-English reasons; titles that are already 4K or HDR, and low copies nobody has played or rated, sit at the bottom. The scoring function is a few lines in `src/main/upgrades.js`, meant to be tuned to taste.

11.2 How the upgrade score works

The score multiplies how far the current copy falls short by how much you care about the title. A higher score means a better candidate; zero or less means not a candidate.

Part	How it is worked out
Shortfall	The resolution gap plus the share of low-bitrate files. Gap: 1080p = 1, 720p = 2, 576p = 3, 480p = 4, SD = 5. The share is low-bitrate files ÷ all files, so it adds up to +1.
Interest	Plex plays (capped at 10, divided by 10) + twice your stars ÷ 5 + half of (online rating – 6) ÷ 4, counted only above 6.
Score	Shortfall × (0.25 + interest), rounded to two decimals.
Always zero	4K titles and anything HDR. Also a title nobody played or rated that has no low-bitrate files: it is not worth the bytes.

Example	Working	Score
A 720p series played 10×, rated 4★, 8.0 online	Shortfall 2. Interest $1 + 1.6 + 0.25 = 2.85$. $2 \times (0.25 + 2.85)$	6.2
A 1080p series, 2 of 10 files low-bitrate, never played, unrated	Shortfall $1 + 0.2 = 1.2$. Interest 0. 1.2×0.25	0.3
An SD movie nobody played or rated	Interest 0 and no low-bitrate files: skipped	0

The reasons shown beside each score are plain words: 720p copy, 3 low-bitrate files, played 8×, you rate it 5★, rated 8.4 online, never played, unrated. The scoring function is a few lines in `src/main/upgrades.js`, meant to be tuned to taste.

PART 3 • REVIEW

12 Ratings and Watch tonight

WHERE Sidebar → Review → Ratings, and Watch tonight	WHO CAN OPEN IT Everyone. Setting stars, notes and tags needs a standard or admin account	IT ANSWERS What do I think of it, and what should we watch tonight?
---	---	---

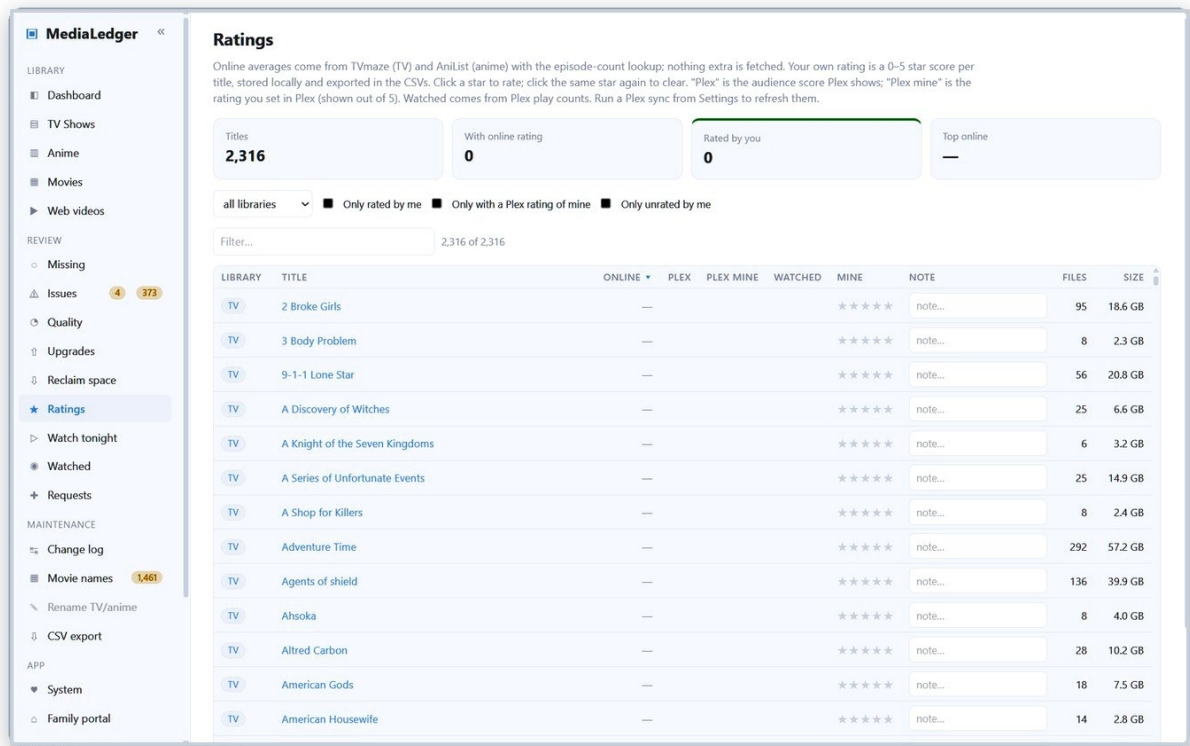


Figure 18. Online averages beside your own stars.

Column	Meaning
Online	The average score from TVmaze or AniList, out of 10, stored when the series was matched. Movies have no online score.
Plex	The audience score Plex shows for the title, once Plex is synced.
Plex mine	The rating you gave the title inside Plex, shown out of 5.
Watched	From Plex play counts: a percentage for series, yes/no for movies.
Mine	Your MediaLedger rating, 0–5 stars. Click to set, click the same star to clear.
Note	Free text saved when you leave the field.

Filters narrow the list to one library, to titles you have rated, to titles you have not, or to titles with a Plex rating of yours. Your stars and notes are exported in the series and movie CSVs.

12.1 Watch tonight

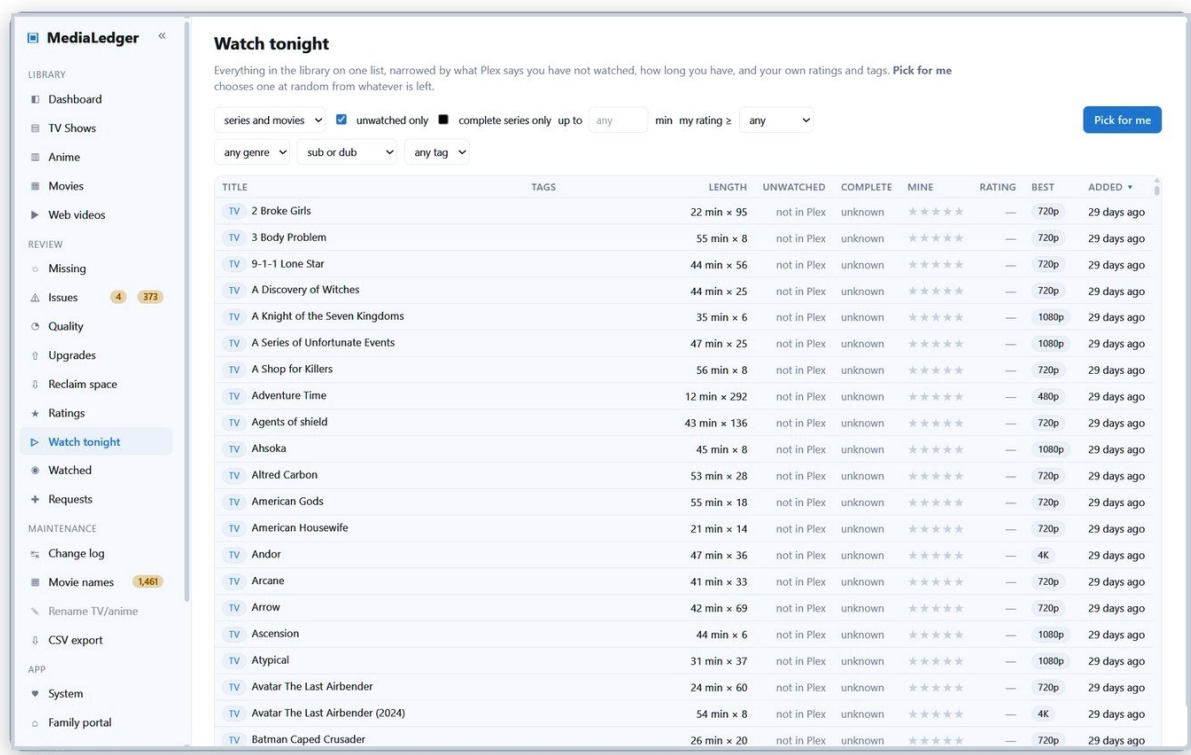


Figure 19. Watch tonight: filters across the top, the Pick for me button and the matching list.

One list across every series and movie, made for the "what shall we watch" moment. Narrow it by kind, **unwatched only** (from Plex play counts; titles Plex does not have stay in), **complete series only** (no missing episodes), a length limit in minutes (episode length for series), your minimum star rating, and the genre, sub/dub and tag dropdowns. **Pick for me** chooses one at random from what is left and shows it in a card with an Open button; **Pick another** rerolls. Your last settings are remembered in the browser.

PART 3 • REVIEW

13 Watched and reclaim space

WHERE

Sidebar → Review → Watched, and Reclaim space

WHO CAN OPEN IT

Standard users and admins

IT ANSWERS

Who watched what, what is next for each person, and what could go?

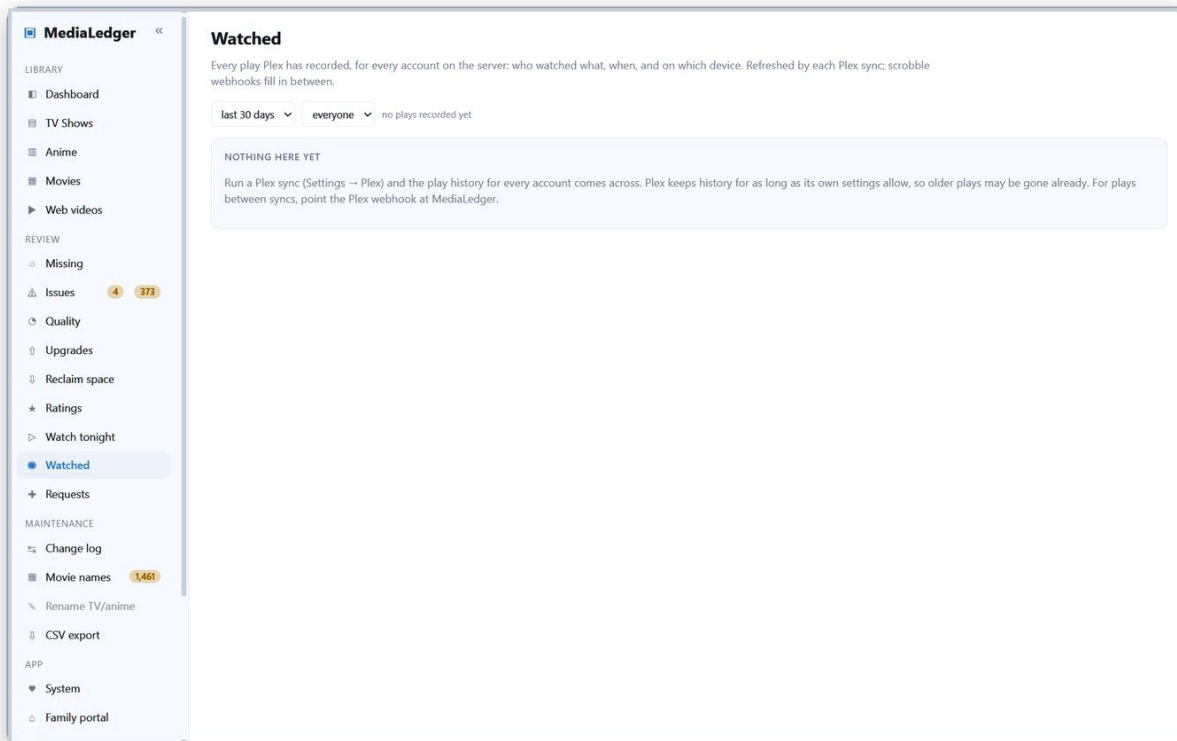


Figure 20. The Watched page: plays by person, library, weekday, hour and device, most watched titles and recent plays.

Watched is the play history of the whole Plex server, one row per play for every account, not just yours. Each Plex sync pulls Plex's own history (Plex keeps it as long as its settings allow), and the scrobble webhook adds plays as they finish ([section 23.3](#)), so the list is current between syncs. Pick a period (7 days to all time) and a person at the top; the tiles count plays, hours (from the files' lengths), distinct titles and people.

The charts break plays down **by person**, **by library**, **by weekday**, **by hour of day** and **by device**, with **plays per day** as a trend. **Most watched series** and **movies** list who watched each and when it was last played. **Recent plays** shows the latest 500, searchable and sortable; a row opens the series or movie when MediaLedger has matched it.

NOTE Plays are attributed by Plex account. Shared accounts show as one person. Adult titles follow the same visibility rule as everywhere else.

13.1 Next up

Next up lists, for each person and each show they watched in the last 60 days, the episode after the furthest one they have seen, with how many are left on disk. **Caught up** means nothing later is on disk. **Gap before it** means the

very next episode is missing and the one shown is the next you actually have, which is a good reason to visit the Missing page (chapter 9).

13.2 Reclaim space

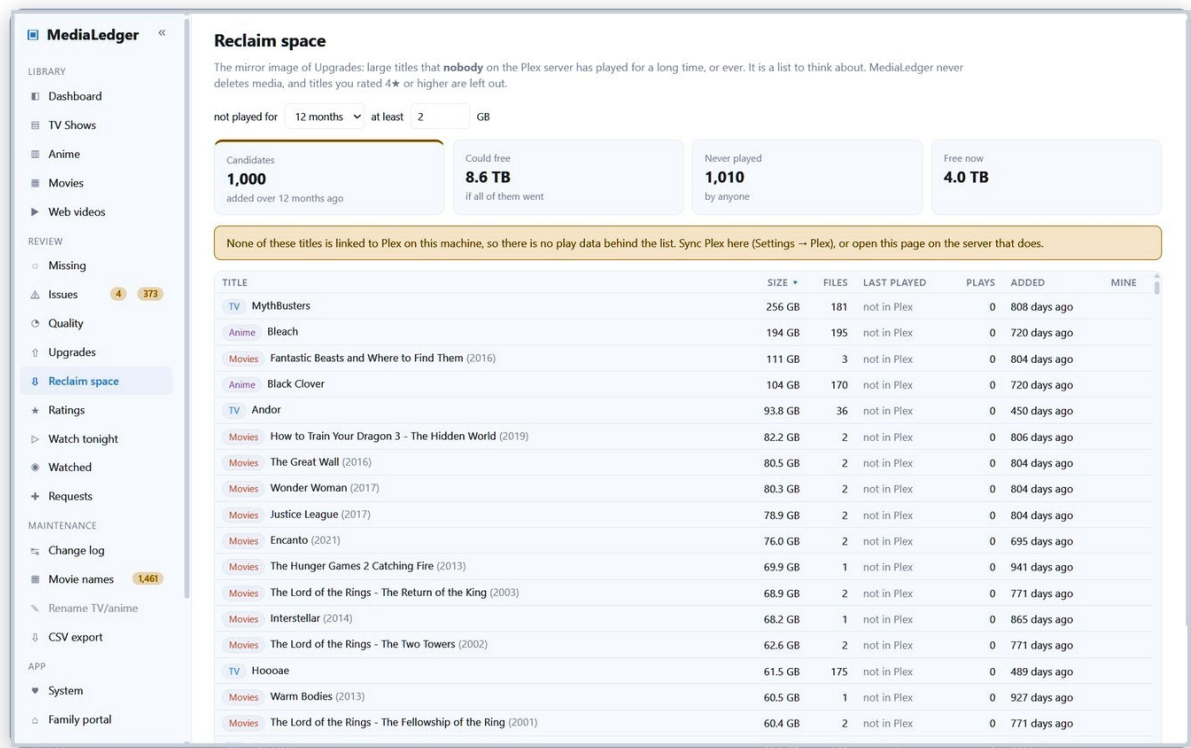


Figure 21. Reclaim space: the largest titles nobody has played in the chosen period.

Reclaim space is the mirror image of Upgrades. It lists large titles that **nobody** on the Plex server has played for the period you choose (6 to 36 months), or ever, and that have been in the library at least that long. "Played" covers every Plex account, from the play history, not just yours. The tiles show how many titles qualify, how much space they hold together, how many were never played, and the free space now. Sort by size and work from the top.

A title is listed when...	
It has been in the library	longer than the period you choose, from 6 to 36 months (12 by default). The date used is the earlier of the day MediaLedger first saw it and the newest file's own date, so a young ledger does not make everything look new.
Nobody has played it	since then: no Plex account, from Plex's own last-viewed date and the play history. A title never played at all is counted separately.
It is big enough	the size threshold (2 GB by default) keeps small titles off the list.
You did not protect it	titles you rated 4★ or higher are left out.

It is a list to think about, nothing more: MediaLedger never deletes media. Titles you rated 4★ or higher are left out, the size threshold keeps small titles off the list, and a title that is not in Plex at all is marked as such rather than as never played.

PART 4

Maintain

The change log, the movie naming engine, TV and anime renaming and CSV export.

IN THIS PART

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One entry per event, with a 30-day activity chart.
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The naming pattern, what is proven and what is a placeholder, and the safety net around every batch.
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The same engine as movies, for episodes. Off by default.
- 17 CSV export**..... 51
Choosing sets, zipping, the files each set writes, and the history.

PART 4 • MAINTAIN

14 Change log

WHERE	WHO CAN OPEN IT	IT ANSWERS
Sidebar → Maintenance → Change log	Standard users and admins	What did each scan find?

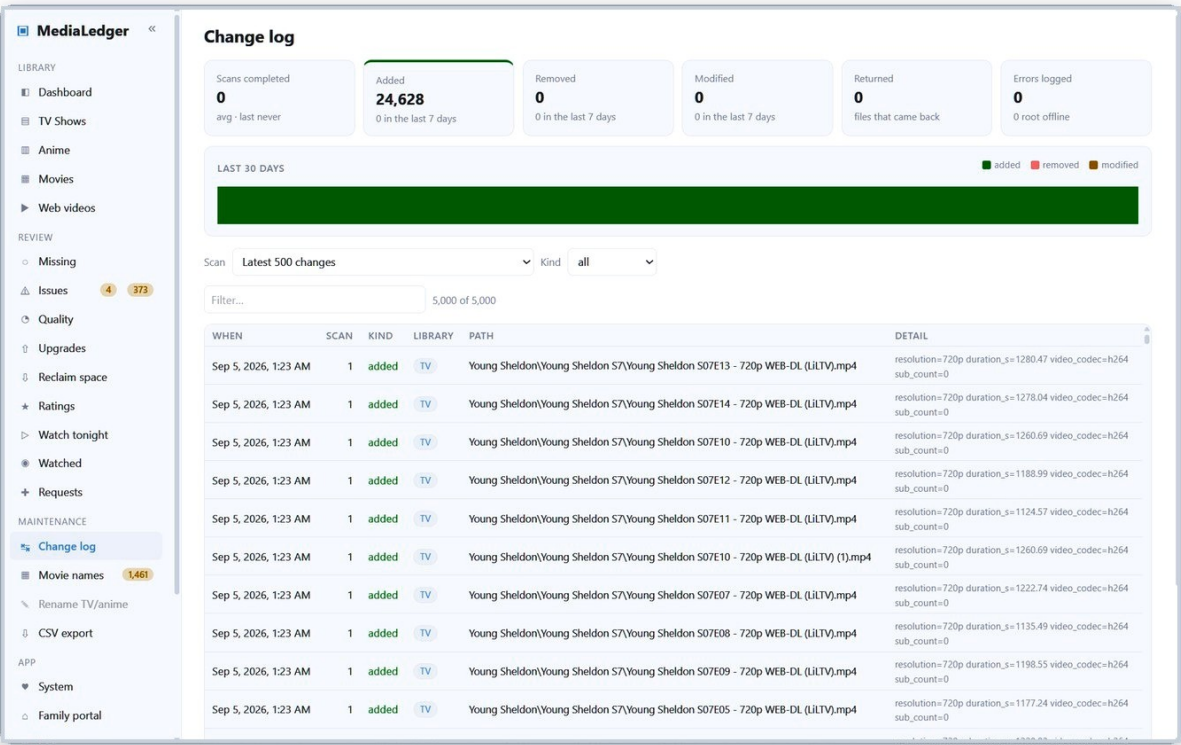


Figure 22. The change log with its 30-day activity chart.

Every scan writes one entry per event. The strip at the top counts totals and the last seven days; the chart shows activity per day for the last 30. Pick a scan in the dropdown to see only its events with its duration, files seen and thread count, or filter by kind.

Kind	Meaning
added	A new file appeared.
removed	A file is no longer on the share. It is flagged missing, never deleted from the ledger.
modified	Size or date changed, so it will be probed again.
returned	A file that was missing came back.
probe_error	ffprobe could not read the file.
root_offline	A root was unreachable and was skipped. Nothing was marked missing.
warning	ffprobe was not found, so files were indexed but not probed.

PART 4 • MAINTAIN

15 Movie names

WHERE

Sidebar → Maintenance → Movie names

WHO CAN OPEN IT

Standard users can look and run dry runs. Only admins rename

IT ANSWERS

What should each movie file be called, and can I rename them safely?

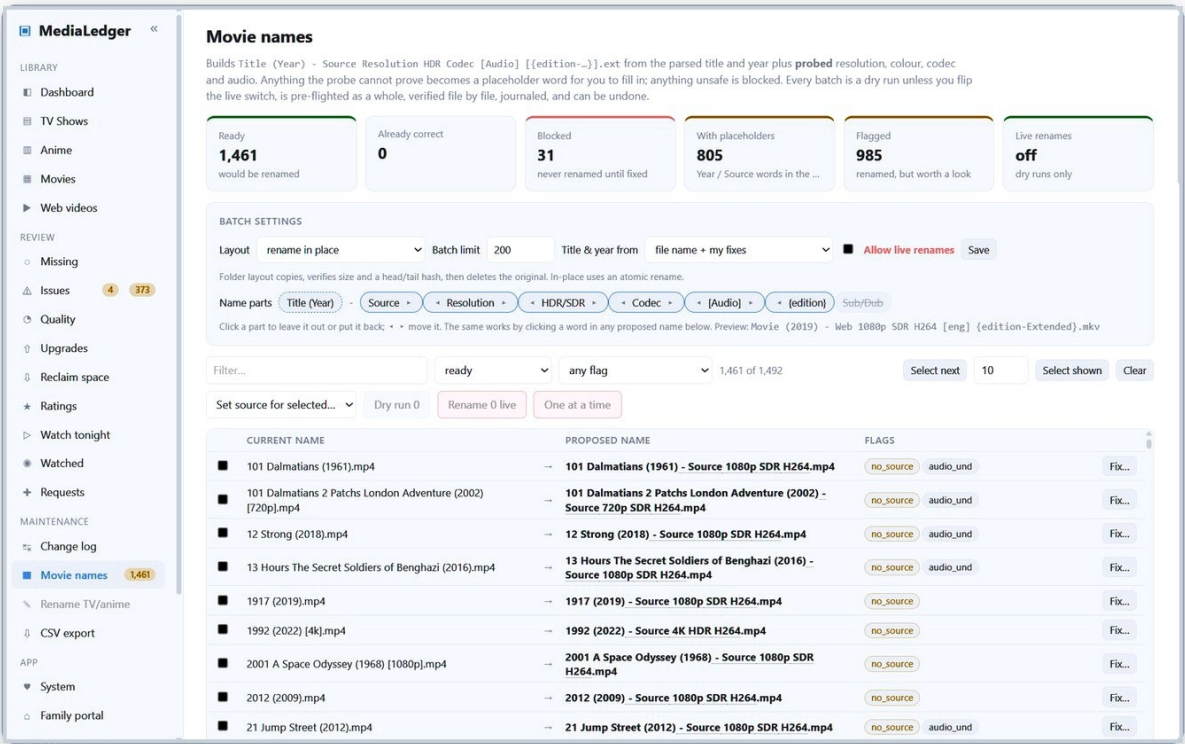


Figure 23. Proposed names with flags. Every batch is a dry run until the live switch is armed.

This tab renames movie files to a consistent pattern built from facts the app can prove. It is the most carefully guarded part of MediaLedger, because it is one of only two places that write to the share.

15.1 The pattern

Title (Year) - Source Resolution HDR Codec [Audio] [{edition-Name}].ext



Examples

A Breed Apart (2025) - Web 1080p SDR H264.mp4

Avatar (2009) - Rip 4K HDR HEVC {edition-Extended Collector's Edition}.mkv

Figure 24. Where each part of a movie name comes from.

A seventh part, **Sub/Dub**, is off unless you switch it on: it writes **Sub**, **Dub** or **Dual** from the probed languages and nothing for a plain English film. Only **Title (Year)** is fixed, because Plex matches on it. Everything after the dash is a **name part** you can leave out or reorder under Batch settings → **Name parts**: click a chip to switch it off (it shows struck through) or on, use the ◀▶ arrows to move it, and watch the preview. The whole list rebuilds at once. The same thing works from the list itself: every word in a proposed name is clickable, so clicking **Source** in any row removes **Source** from every name, the way editing one dimension in a CAD sketch updates the whole part. If you do not like **Web / Rip** in your names, click it once and it is gone; the **no_source** flag disappears with it.

Examples: A Breed Apart (2025) - Web 1080p SDR H264.mp4 and Avatar (2009) - Rip 4K HDR HEVC {edition-Extended Collector's Edition}.mkv.

Token	Where it comes from	When unknown
Title, Year	The parser and your manual fixes, or the Plex match when the truth source is set to Plex.	Year becomes the word Year; the file is flagged.
Source	Web when the old name carries a LiTV or WEB marker; Rip for BRrip, BluRay, BDRip, Remux or DVD; or whatever you set in the Fix dialog or with bulk source.	The word Source; the file is flagged.
Resolution	ffprobe pixel size: 4K, 1440p, 1080p, 720p, 576p, 480p or SD.	Blocked.
HDR / SDR	ffprobe colour transfer: HDR10, Dolby Vision and HLG all become HDR.	SDR.
Codec	ffprobe video codec: H264, HEVC, AV1, MPEG4, VC1...	Blocked.
Audio	ffprobe audio languages, only when they are not plain English, e.g. ENG+JPN.	Omitted; flagged if undefined.
Edition	Edition words found in the old name, in Plex's {edition-...} form.	Omitted.

The title keeps your existing words. Colons and question marks are dropped, slashes become dashes, other characters Windows forbids are removed, and trailing dots and spaces are trimmed. Nothing is re-capitalised.

15.2 Ready, flagged, blocked

- **Ready** – the proposed name differs from the current one and every check passes.

- **Flagged** – ready, but worth a look. Flags include `no_source`, `no_year`, `res_mismatch` (the old name claimed a resolution the probe disagrees with; the probe wins), `hdr_claimed_but_sdr`, `audio_und`, `plex_title_differs` and `plex_unlinked`. The tab explains each flag under "What the flags mean".
- **Blocked** – never renamed: no successful probe, no title, or a **collision** where two or more files would receive the identical name.

15.3 Batch settings

Setting	Effect
Layout	Rename in place changes only the file name (an atomic rename). Move into "Title (Year)" folders creates Plex's preferred folder layout by copying the file, verifying its size and a head-and-tail hash, then deleting the original.
Batch limit	The most files one live batch may contain. Forces review in chunks.
Title & year from	File name + my fixes (default) or Plex match . With Plex, unlinked files fall back to the file name and are flagged.
Allow live renames	The arming switch. Until it is on, only dry runs are possible.

15.4 The safety net

Renaming is one of only two places MediaLedger writes to your share, so it is built to be checked at every step and undone afterwards. Nothing is ever overwritten, and every attempt is journaled.

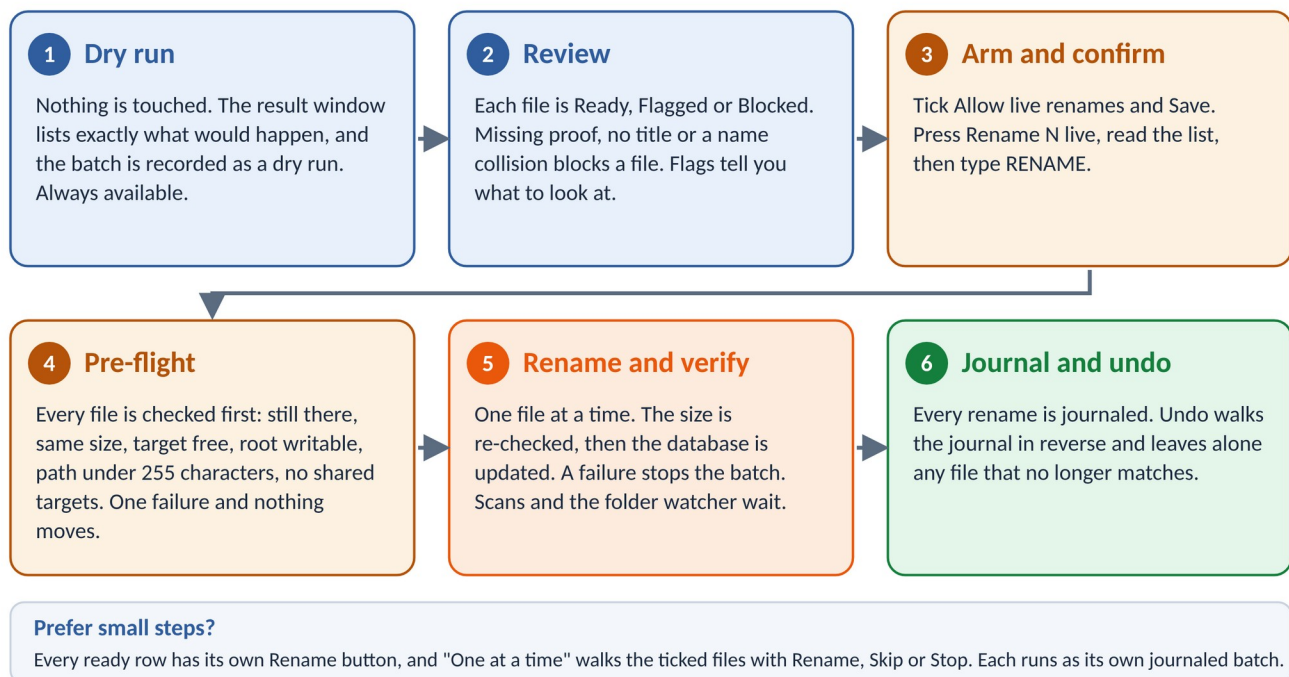


Figure 25. The path every live batch takes. Movies and TV/anime use the same engine.

15.5 Running a batch

1. Filter and tick the files you want. **Select next** with the number beside it clears the selection and ticks only that many of the listed files (10 unless you change it; the number is remembered), so a run never covers more than you meant. After the batch those files leave the list and **Select next** picks up the following group. **Select shown** ticks every ready file currently listed and asks first when that is more than 50.
2. Press **Dry run N**. Nothing is touched; the result window lists exactly what would happen and the batch is recorded as a dry run in the Batches table.
3. When satisfied, tick **Allow live renames**, press **Save**, then **Rename N live**.
4. A confirmation lists the renames and asks you to type **RENAME**.
5. **Pre-flight** then checks every file: it still exists, its size matches the last scan, the target name is free, the root is writable, the path is under 255 characters and no two files share a target. If any check fails, the whole batch is aborted and nothing is renamed.
6. Each rename is followed by a re-check of the file's size before the database is updated. A failure stops the batch; earlier renames stand and are journaled.

While a live batch runs, scans and the folder watcher wait.

One at a time

If a big batch feels like too much trust, do not use it. Every ready row has a **Rename** button that renames just that file, and **One at a time (N)** walks through the ticked files in a dialog that shows the current and proposed name and asks **Rename this file**, **Skip** or **Stop** for each. Every confirmed file runs as its own one-file batch through the same pre-flight, verification and journal, so it appears in the Batches table and can be undone on its own. The live switch must still be on.

15.6 Undo

Every live batch is listed under **Batches** with an **Undo** button. Undo walks the journal in reverse; for each file it confirms the renamed file still exists with the recorded size and that the original name is free, then renames it back. Files that fail the check are left alone and reported, and the batch shows as "partially undone". **Items** shows the per-file journal of any batch.

15.7 Bulk source, collisions, placeholders

- **Set source for selected...** applies Web, Rip or "clear" to every ticked file (or to everything shown when nothing is ticked). It only records a fix; nothing is renamed.
- **Name collisions** groups the blocked pairs with their sizes. **Keep this, ignore others** marks the rest ignored (still on disk) so the kept file can be renamed; alternatively give one file a distinguishing edition via Fix...
- **Placeholders on disk** lists files already renamed with **(Year)** or **Source** in their name. Plex will not match a **(Year)** file, so fill the value with Fix... and rename again.

CAUTION Before the first large live batch, run one small batch of 20 or 30 files in place, check the result in Explorer and in Plex, then undo it. That proves the whole path on your NAS before you trust it with the full library.

PART 4 • MAINTAIN

16 Rename TV and anime

WHERE Sidebar → Maintenance → Rename TV/anime	WHO CAN OPEN IT Standard users can dry-run. Only admins rename. Off until switched on	IT ANSWERS How do I give episodes Plex-standard names?
---	---	--

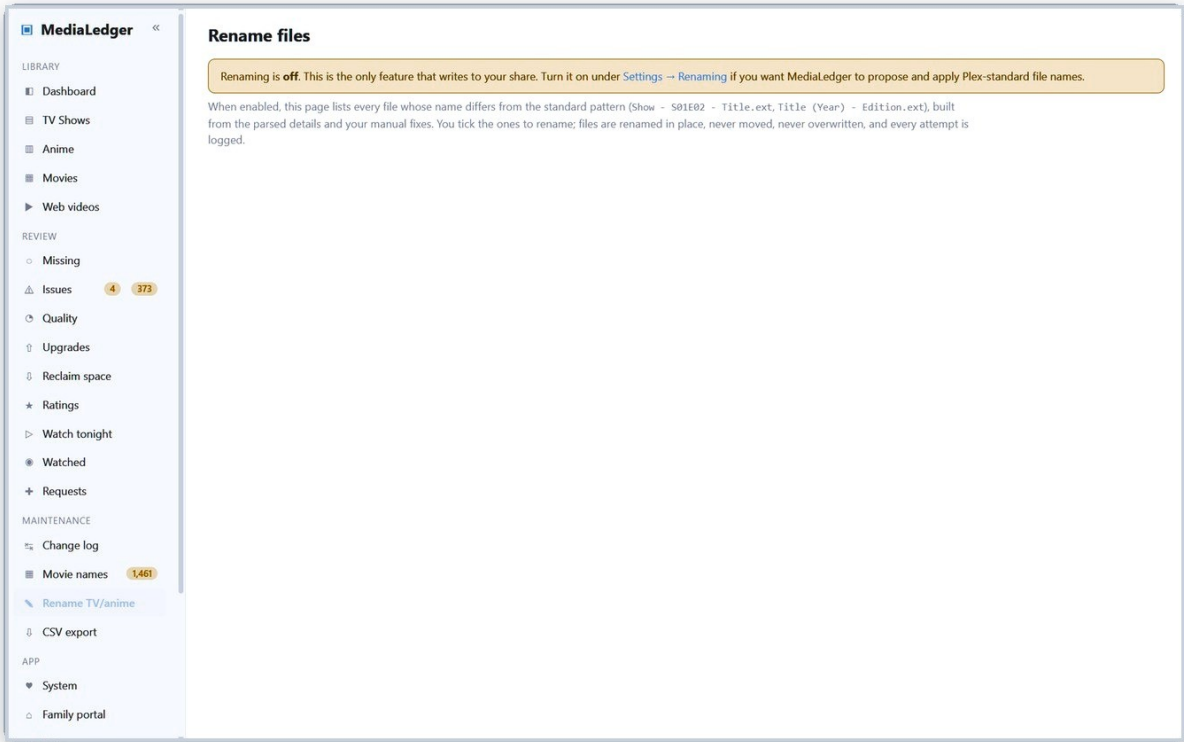


Figure 26. The episode rename tool, shown here while disabled.

Episodes use the same engine as movies. Every run is a batch that is pre-flighted as a whole (file still there, size unchanged, target free, root writable), each rename verified, journaled in the **Batches** table and undoable from it (section 15.6). **Dry run N** records what would happen without touching anything.

The tool is **off** until you enable it under Settings → Renaming. When on, it lists every episode file whose name differs from `Show - S01E02 - Title.ext`, built from the parsed details and your fixes. Files are renamed in place, never moved or overwritten, and every attempt is logged in the History table.

Name part	State	Written as
Show - S01E02	Fixed	The show, season and episode.
Episode title	On by default	- Title after the number.
Resolution, Codec, Sub/Dub	Opt-in	In square brackets, for example [1080p].

The one-at-a-time scheme works as on the movie page: a **Rename** button per row, and **One at a time (N)** for the ticked files, each confirmed separately. **Select next** takes a set number of episodes at a time: press it, check the ticked rows, run the batch, repeat. It is the comfortable middle between one file at a time and everything shown.

TIP Fix anything wrong under **Issues** → **Problems** first ([section 10.1](#)). The proposal is only as good as the parse.

PART 4 • MAINTAIN

17 CSV export

WHERE Sidebar → Maintenance → CSV export	WHO CAN OPEN IT Standard users and admins. Admins download the files on the web server	IT ANSWERS How do I get the ledger into a spreadsheet?
--	---	--

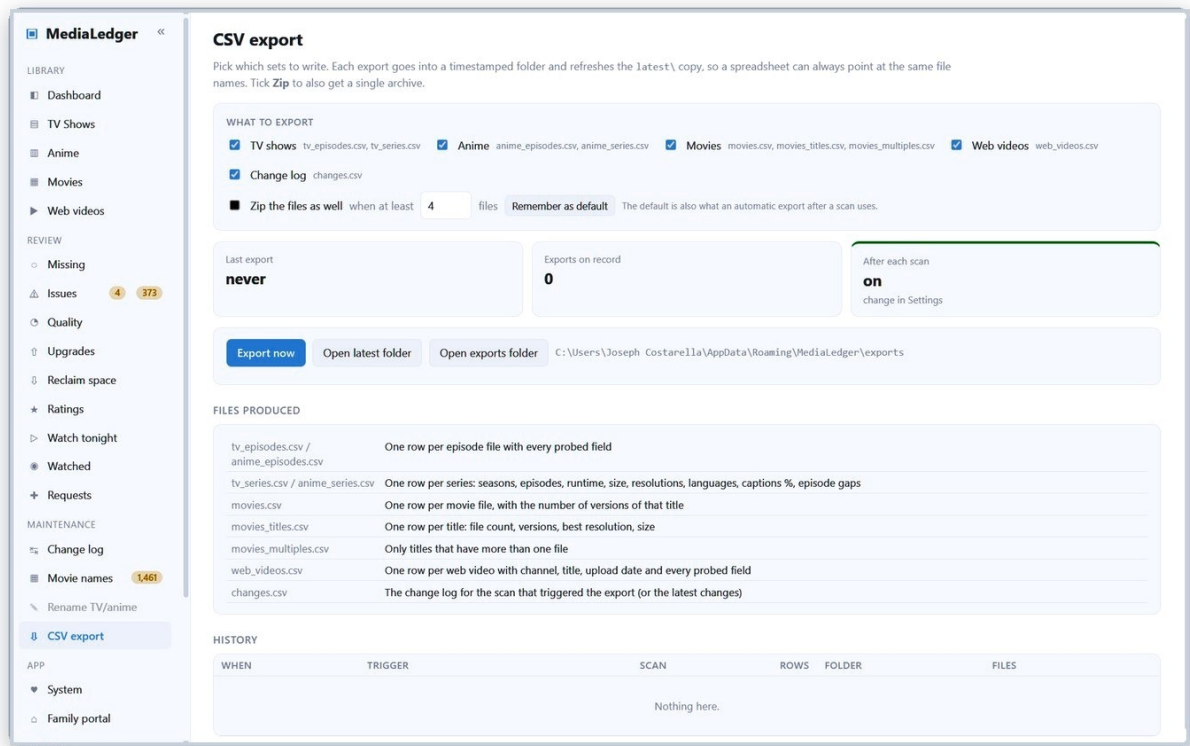


Figure 27. The export page: pick the sets, optionally zip them, export, and browse the history.

Every export writes a set of CSV files into `exports\<timestamp>\` and refreshes the `latest\` copy, so a spreadsheet can always point at the same file names. Exports run automatically after each scan (Settings → CSV export) or on demand with **Export now**. The history table lists past exports with the trigger, the scan, row counts and the files produced.

17.1 Choosing what to export

- **What to export** lists the five sets: TV shows, Anime, Movies, Web videos and Change log. Tick the ones you need; the file names each set produces are shown beside it.
- **Zip the files as well** adds `medialedger-<timestamp>.zip` next to the CSVs and refreshes `latest\medialedger-latest.zip`. The archive is written by MediaLedger itself; no extra software is needed. The **when at least N files** box beside it skips the zip for small exports: with the default of 4, ticking only the change log gives a bare CSV, ticking everything gives the zip too.
- **Remember as default** saves the current ticks. The default is also what the automatic export after a scan uses.

- On the web server the history table links every CSV and zip for download in the browser (admin session). On the desktop the **Open latest folder** and **Open exports folder** buttons open the files in Explorer.

Setting	Default
Sets	TV shows, Anime, Movies, Web videos and Change log
Zip the files as well	Off. When on, the zip is made when at least 4 files were written
Export after every scan	On
Output folder	Empty, which means exports\ in the data folder
Adult roots	Excluded unless allowed in Settings → Adult content

17.2 The files

File	Contents
tv_episodes.csv / anime_episodes.csv	One row per episode file with every parsed and probed field.
tv_series.csv / anime_series.csv	One row per series: seasons, episodes, runtime, size, resolutions, codecs, languages, captions %, episode gaps, metadata source, expected and missing episodes with the exact list, online rating, your rating and note, genres, sub/dub, your tags, mixed-resolution and low-bitrate flags.
movies.csv	One row per movie file, with the number of versions of that title.
movies_titles.csv	One row per title: file count, versions, best resolution, size, languages, captions, your rating and note, genres (from Plex), sub/dub, your tags.
movies_multiples.csv	Only titles with more than one file.
web_videos.csv	One row per web video: channel, title, upload date, video id and every probed field.
changes.csv	The change log for the scan that triggered the export.

Adult roots are excluded unless allowed in Settings. Ignored files are always excluded.

PART 5

Share and automate

Requests, the family portal, notifications and the jobs that run by themselves.

IN THIS PART

- 18 Media requests**..... 54
A shared wish list, plus a phone-sized page for guests.
- 19 Family portal**..... 56
Invites, what visitors see, addresses, optional admin sign-in from outside, and the health check.
- 20 Notifications and Home Assistant**..... 59
A webhook or e-mail for six kinds of event, and a read-only status JSON for Home Assistant.
- 21 Schedules, backup and restore**..... 61
Every automatic job in one table, how scans are scheduled, the nightly backup and restoring it.

PART 5 • SHARE AND AUTOMATE

18 Media requests

WHERE Sidebar → Review → Requests, and the phone page	WHO CAN OPEN IT Everyone can file. Only admins change a status	IT ANSWERS How do people ask for a title, and how do I see what is waiting?
---	--	---

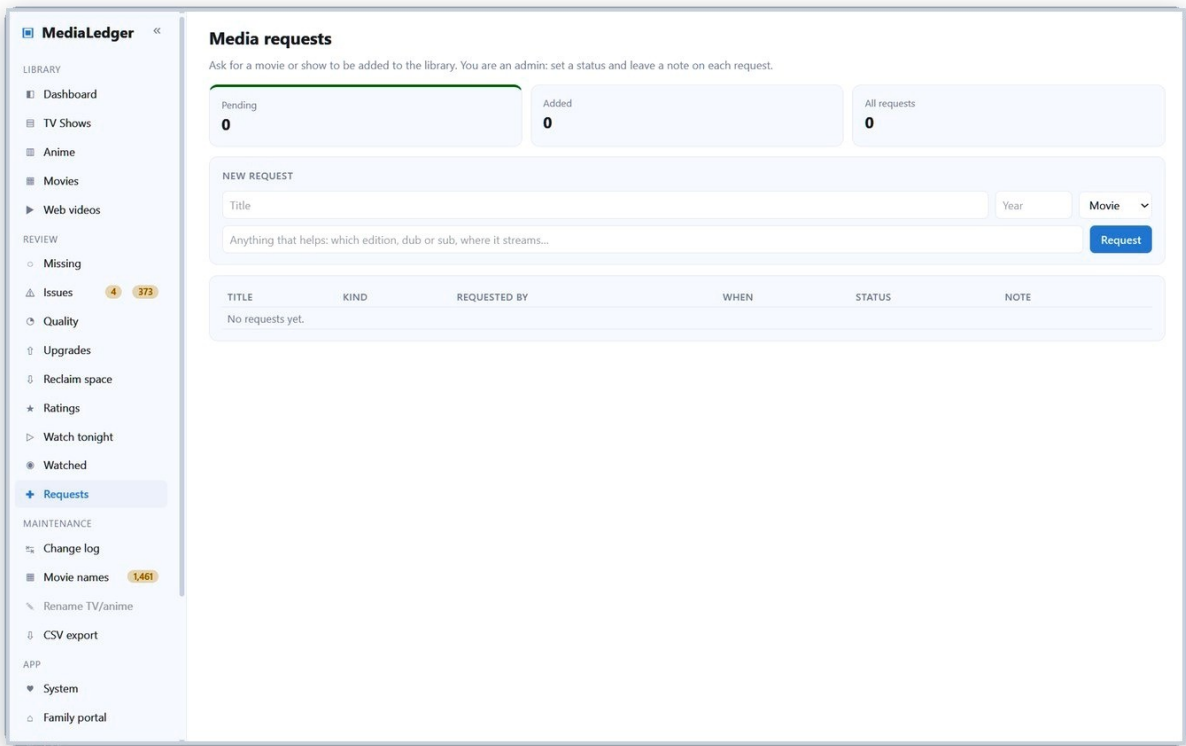


Figure 28. The Requests page as an admin sees it.

Requests is a wish list for the library. Anyone who can open the app may ask for a title: enter the name, the year if you know it, the kind (Movie, TV show, Anime, Other) and a note that helps whoever fulfils it, such as the edition, dub or sub, or where it streams. Guests on the web server also type their name, since they are not signed in.

Every request has a status: **Pending** when filed, then **Approved**, **Added** or **Declined**. Only an admin changes the status, and only an admin sees the extra column with the status dropdown, the  button for a note back to the requester, and **X** to delete the request. The tiles at the top count pending, added and all requests; on the web server the sidebar shows the pending count to admins so new requests are noticed.

NOTE Requests do not download anything and do not talk to Plex. They are a shared list between the people who use the library and the person who maintains it.

18.1 The phone page

<http://medialedger.local:8080/request> is a phone-sized version of the form with nothing else on it. The guest QR code on the Security tab points there, so a visitor scans, types a title and is done. Guests give their name

once and the phone remembers it. People you invite to the family portal ([chapter 19](#)) have their own request form there.

PART 5 • SHARE AND AUTOMATE

19 Family portal

WHERE Sidebar → App → Family portal	WHO CAN OPEN IT Admins manage it. Invited people use it	IT ANSWERS How do I let family browse and request titles without giving them the app?
---	---	---

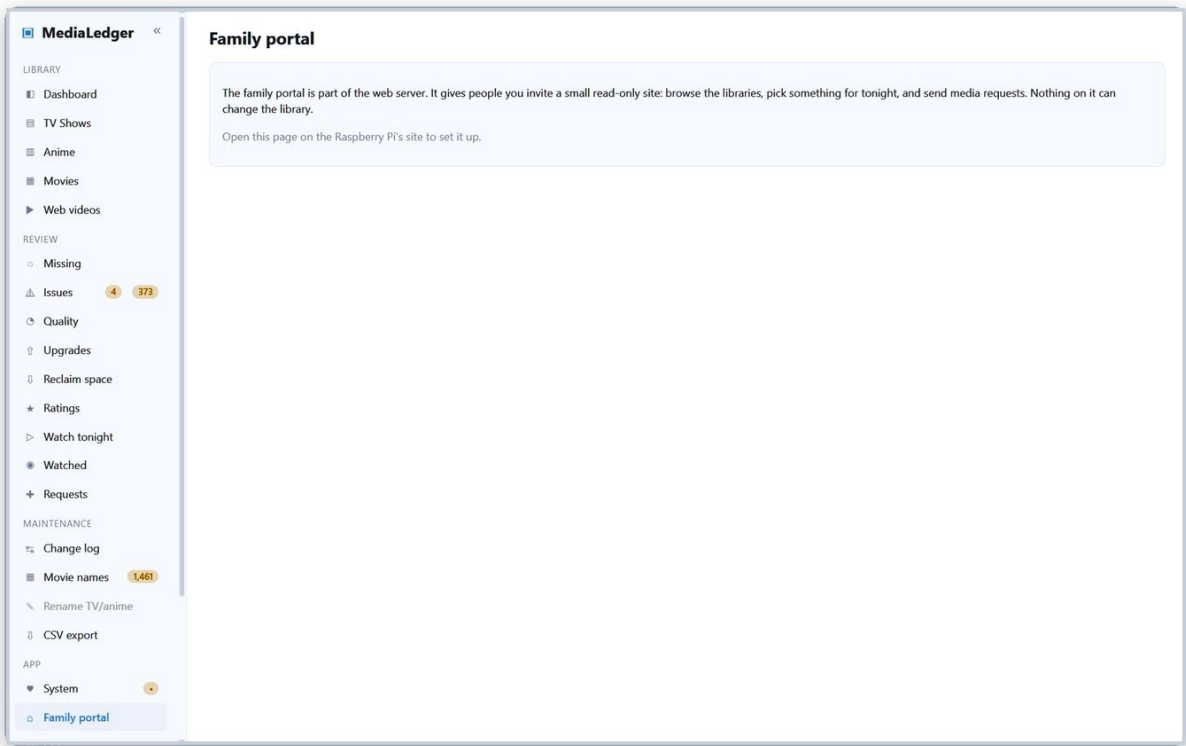


Figure 29. The Family portal page: invites, addresses and the switch.

The **family portal** is a second front door for people you invite. It lets them browse the three libraries, use **What to watch tonight**, and send **media requests**, from a small site made for phones. It is a separate listener on its own port with its own short list of operations: sign-in, settings, scans, renames, exports and logs do not exist there, so there is nothing to break into. Every reply is built from a fixed list of fields, so file paths, the names of your NAS and shares, Plex keys, watch history and your notes never leave the server, and adult titles are never included.

19.1 Portal and web app compared

	Web app	Family portal
Listener	Port 8080, the whole program	Port 8090, its own short list of operations
Reachable from	Private addresses only	Wherever you publish it. It binds to this machine only, so a proxy or Funnel sits in front
Sign-in	Username and password, optional two-factor code. Sessions last 30 days	An invite link. The device is remembered for 180 days (about six months)

	Web app	Family portal
Sees	Everything its role allows	Titles, seasons, what is missing, genres, posters. No paths, no ids, no Plex fields, no adult titles
Can change	Whatever its role allows	Only its own requests. Six an hour and 25 a day per person

19.2 Invites

Invites. On the **Family portal** page type a name, choose an expiry and what the person may do, and press **Create invite**. You get a link and a QR code, shown once: send it then. Opening the link on a phone remembers that device for six months; the link itself is only needed again on a new device. The table shows each person, when they were last seen, how many devices and requests, and gives **New link** (the old link and its devices stop working) and **Revoke**. Only a fingerprint of each link is stored, so a copied database cannot recreate one.

Permission	Lets the person
Browse	Search the library and open title pages
Tonight	Use What to watch tonight and Pick for me
Request	File media requests. The form checks the library first

19.3 What they see

What they see. Library: search, genre, quality and sub or dub filters, and a page per title with the seasons and episodes on hand and what is missing. Tonight: the same filters and **Pick for me**; "not watched yet" follows the household Plex account. Requests: the form checks the library first and shows *already in the library* before sending; each person sees their own requests with your status and reply. Requests arrive in your Requests tab marked **family: Name**, with the usual notification. Filing is limited to six requests an hour and 25 a day per person.

19.4 Addresses

Addresses. Give the portal a **public address** (what people away from home open, see [section 27.4](#)) and optionally a **home address** served by Caddy. Invite links are built from them. Leave the portal reachable from **this machine only** when Caddy or Tailscale sits in front. Switching the portal **off** closes it for everyone at once and keeps the invites.

19.5 Admin sign-in from outside

You, from outside. Tick **Admin sign-in from outside** and press Save, and the portal gains a page of its own for you at `<public address>/admin`: sign in with your admin account (the same password, lockout and two-factor code as the web app; standard accounts are refused) and handle the day-to-day without being on the home network. Three tabs: **Requests** (approve, mark added, decline, reply, delete), **Invites** (create with the link and QR code, new link, revoke, delete) and **Status** (files, free space, waiting requests, missing episodes, the address checks, the Schedules table with **Run now** for the scan, Plex sync, metadata, posters, backup, snapshot, summary and the address check, and the two portal options). Nothing else is reachable there: settings, renames, folders, files and logs stay at home, and the switch itself, the port and the addresses can only be changed on this page. Sessions last twelve hours or an

hour idle; switching the option off ends them all. Turn on two-factor codes under Security first: this page is on the internet.

CAUTION This page is on the internet. Turn on two-factor codes under Security first ([section 25.4](#)).

19.6 Is it working?

Is it working? The server tests its own addresses the way a visitor would: it looks the name up, connects, checks the certificate and asks the portal to identify itself, so a name that points somewhere else does not pass. It does this every 15 minutes (set the interval at the bottom of the page; 0 turns it off) and when you press **Check the addresses now**. The two tiles show **working** with the response time, when it was last checked, how long it has been up or down, and the days left on the certificate. One failed check turns a tile amber; two in a row turn it red, put a notice on the Dashboard, write the log and, when **family portal not reachable** is ticked under Notifications, send a message, with another when it comes back. The check is also a row under Settings → Schedules. A home address served by Caddy uses Caddy's own certificates, which the check accepts without verifying, and says so.

PART 5 • SHARE AND AUTOMATE

20 Notifications and Home Assistant

WHERE

Settings → Automation → Notifications

WHO CAN OPEN IT

Admins

IT ANSWERS

How do I get told when something happens?

MediaLedger can tell you about things without you opening the app. Two channels exist and you can use both at once. A **webhook URL** receives a small JSON body per event: it works with a Home Assistant webhook trigger, ntfy, Discord and anything similar. **E-mail** goes through any ordinary mailbox using a dependency-free SMTP client; for Gmail use an app password.

20.1 Events

Event	Sent when	On by default
New media request	Someone files a request, from the app, the phone page or the family portal	Yes
Daily summary	Once a day at the time you choose (08:00 by default): files and free space, pending requests, missing episodes, what airs this week, series that finished airing but are incomplete	Yes
Backup failed	The nightly backup could not write	Yes
A scheduled job is overdue	A job has not run for twice its interval. At most once a day	Yes
Family portal not reachable	Two address checks in a row failed, and again when it recovers	Yes
Episodes airing	Adds the airing list to the daily summary	Yes

Send a test saves the settings and sends one message to every channel you have set up.

20.2 What a webhook receives

One JSON `POST` per event with the fields below. Some events add their own, for example the daily summary adds `pending_requests`, `airing_this_week` and `free_bytes`.

EXAMPLE BODY

```
{
  "app": "MediaLedger",
  "event": "dailySummary",
  "title": "Daily summary: 2 requests, 5 airing this week",
  "message": "Files: 24,628 (12.10 TB)\nFree on the share: 4.02 TB ...",
  "pending_requests": 2,
  "airing_this_week": 5,
  "free_bytes": 4020000000000,
  "at": "2026-10-05T08:00:00.000Z"
}
```

20.3 Home Assistant status

Home Assistant status is a read-only JSON summary at a URL with its own key, shown in the same Settings section. **New key** rotates it. The request needs no session, and the LAN-only rule still applies. Add it as a RESTful sensor and pick values with a template such as `{{ value_json.pending_requests }}`.

HOME ASSISTANT CONFIGURATION.YAML

```
rest:
- resource: http://medialedger.local:8080/api/status?key=YOUR_KEY
  scan_interval: 300
  sensor:
    - name: "MediaLedger pending requests"
      value_template: "{{ value_json.pending_requests }}"
    - name: "MediaLedger missing episodes"
      value_template: "{{ value_json.missing_episodes }}"
```

Field	Meaning
files, bytes	Files and total size, adult excluded
free_bytes, months_left	Free space on the roots, and months until full at the current growth rate
pending_requests	Requests waiting for a decision
missing_episodes	Episodes you lack, after your collecting choices
airing_this_week, next_airing	Count, and the next one as <code>show</code> , <code>date</code> , <code>episode</code>
scanning, last_scan	Whether a scan is running, and the last one as <code>finished</code> , <code>status</code> , <code>added</code> , <code>removed</code>
app, at	The word MediaLedger, and when the reply was made

PART 5 • SHARE AND AUTOMATE

21 Schedules, backup and restore

WHERE

Settings → Automation → Schedules, and Backup and data

WHO CAN OPEN IT

Admins

IT ANSWERS

What runs by itself, and how do I back up and restore?

21.1 The Schedules table

Settings → **Schedules** lists every job MediaLedger runs on its own, in one table: the library scan, the Plex sync, episode counts and airing dates, the backup, the daily snapshot and the daily summary. Each row shows when it runs, the last run and its result, the next run, and a **Run now** button. The timing controls sit underneath: the **Plex sync** interval (a timer keeps the Watched tab and play counts current on days without a scan; 0 means only after scans), the backup time and how many copies to keep, the snapshot time and the summary time. The scan timers and the Windows scheduled task follow.

Job	Default
Library scan: in-app timer	Off. When on, every 24 hours while the window is open
Library scan: Windows Task Scheduler	Off. When installed, daily at 03:00 as “MediaLedger Scan”
Plex sync	Every 6 hours when Plex is switched on, and after every scan
Expected episodes and airing dates	On. Airing series are re-checked every 14 days
Daily snapshot (trend cards)	03:05, when no scan has taken one that day
Daily summary	08:00, when you have a channel set up
Nightly backup	Off. When on, 03:30, keeping the newest 7
Folder watch	Off. When on, a scan starts after the folder has been quiet for 90 seconds
Root reachability check	Every 5 minutes. A Settings badge and a notice appear when a root disappears

21.2 Overdue jobs

A job is **overdue** when it is enabled but has not run for twice its interval (48 hours for the daily ones), or when the last backup failed. Overdue jobs get a red badge in the table, a notice at the top of the Dashboard for admins, a line in the log and, once a day, a notification if a **scheduled job is overdue** is ticked under Notifications.

21.3 Scheduled scans on Windows

- **In-app timer** – scans every N hours while the window is open.
- **Windows Task Scheduler** – installs a daily task that launches MediaLedger with `--scan`, which scans, exports and exits even when the app is closed. If the app is already open, the open window runs the scan instead. Re-install the task after upgrading so it points at the current program.

NOTE --scan with the app already open hands the request to the open window. With nothing open it runs headless: scan, export, quit. If you open the window mid-scan the headless instance opens it for you and keeps going.

21.4 Backup

Database path and size, schema version, counts, and buttons to back up now, open the backups folder, the data folder and the log. **Nightly backup to a folder** copies the database to a folder of your choice once a day at the time you set, dated, keeping the newest N. Point it at the NAS: the database holds every fix, rating, tag, match and the whole change log, and this is the only copy that survives a dead SD card or PC. Each backup is a **set** with one date stamp: the database, `settings.json` and, on the web server, `web.json` (accounts, password hashes and 2FA secrets, so keep the folder private). Sets are pruned together. **Back up there now** tests the folder.

21.5 Restore

Restore... lists the sets in the folder. Pick one and choose whether to bring back the settings and the web accounts as well as the library data. The backup is opened read-only and checked (integrity, schema no newer than this build) before anything happens; then MediaLedger restarts and swaps the files in before opening them. The files it replaces are not deleted: they are moved into a `pre-restore-<date>` folder next to the database. On the web server a restore asks for your password again and is written to the security log.

TIP Point the backup at a folder on the NAS. It is the only copy that survives a dead SD card or PC, and it holds every fix, rating, tag, match and the whole change log.

PART 6

Settings and connections

Every setting, grouped the way the Settings page groups them, and the Plex link.

IN THIS PART

- 22 Settings reference**..... 64
The seven groups on the Settings page, with the defaults you start from.
- 23 Plex integration**..... 68
Connecting, what a sync stores, and the optional webhook.

PART 6 • SETTINGS AND CONNECTIONS

22 Settings reference

WHERE	WHO CAN OPEN IT	IT ANSWERS
Sidebar → App → Settings	Admins	Where is that setting, and what is its default?

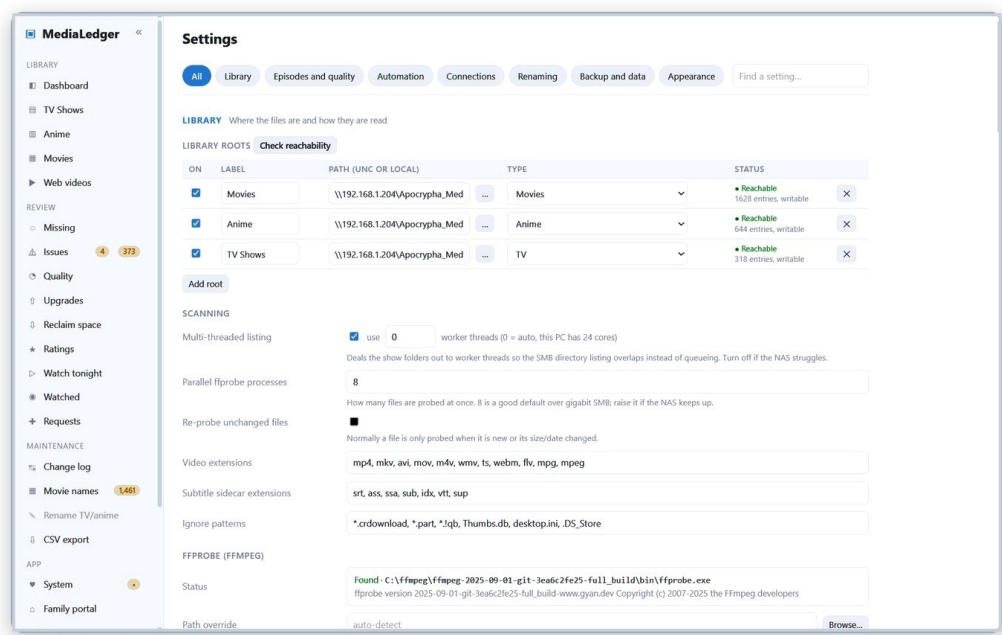


Figure 30. The top of the Settings page, with its group tabs and the Find a setting box.

The page is divided into seven groups, each with a tab at the top: **Library** (roots, scanning, ffprobe, folder watch, adult content), **Episodes and quality** (expected episodes, quality thresholds, posters), **Automation** (schedules, notifications, updates), **Connections** (Plex, CSV export), **Renaming**, **Backup and data**, and **Appearance**. **All** shows everything under its group headings. **Find a setting** filters the sections by any word in them, across all groups. One **Save settings** button at the bottom saves the whole page whichever tab is showing.

Group	Holds
Library	Library roots, Network shares, Scanning, ffprobe, Folder watch, Adult content
Episodes and quality	Expected episodes, Quality thresholds, Posters
Automation	Schedules, Notifications, Updates
Connections	Plex, CSV export
Renaming	The TV/anime and movie rename switches and name parts
Backup and data	Database facts, nightly backup, restore
Appearance	Colour theme

22.1 Library

Library roots

One row per folder: on/off, label, path (UNC or local; the ... button browses), and type: TV, Anime, Movies, Web videos, or Adult (auto-detect anime / TV / movie). Roots with the same type may repeat. Press **Save settings** after editing.

The defaults are three roots, **Movies**, **Anime** and **TV Shows**, all enabled. On Windows they point at a UNC path and on the Pi at `/mnt/media/...`; change them to match your share.

Network shares (Raspberry Pi only)

On the Pi the media lives on a file server, and the server must be connected to it before any folder can be picked.

Connect a network share... asks for the server's address (an IP address or a name), a username and a password, then lists the shares that server offers. Tick the ones that hold media and press **Connect**: each appears as a folder under `/mnt/medialedger/`, shown in the table with its status, and as a button at the top of the folder picker.

Remove disconnects one; nothing on the file server changes. The sign-in is stored on the Pi in a file only root can read, so shares come back after a restart; it is never written to MediaLedger's settings. Connecting and removing ask for your password again. On the Windows desktop app there is nothing to connect: browse straight to the folders, or type a path like `\\server\share\Movies`.

The welcome guide opens by itself on a new install and from **Open the setup guide** here. It walks through four steps on one page: connect storage, pick the TV, Anime and Movies folders (**Find them for me** looks inside the connected shares for folders with the usual names), check that ffprobe is present and every folder can be read, and start the first scan.

Scanning

- **Multi-threaded listing** deals show folders out to worker threads so directory listing over SMB overlaps. 0 threads means automatic (CPU count minus one). Turn off if the NAS struggles.
- **Parallel ffprobe processes** – how many files are probed at once; 8 suits gigabit SMB.
- **Re-probe unchanged files** – force ffprobe on everything next scan.
- **Video extensions, Subtitle sidecar extensions, Ignore patterns** – what counts as media, what counts as a caption sidecar, and glob patterns to skip.

Setting	Default
Multi-threaded listing	On. Threads: automatic, which is CPU count – 1, at least 2 and at most 32
Parallel ffprobe processes	8
Re-probe unchanged files	Off
Video extensions	mp4 mkv avi mov m4v wmv ts webm flv mpg mpeg
Subtitle sidecar extensions	srt ass ssa sub idx vtt sup
Ignore patterns	*.crdownload *.part *!.qb Thumbs.db desktop.ini .DS_Store

ffprobe (ffmpeg)

Shows where ffprobe was found and its version. A path override and a **Download** button fetch the latest static build into the data folder.

Folder watch

Uses Windows change notifications on each root and starts a scan once the folder has been quiet for the settle time. A copy in progress keeps pushing the timer back.

Off by default. The settle time is 90 seconds.

Adult content

Whether adult roots appear in CSVs, and the fallback subtype for files that match no pattern.

22.2 Episodes and quality

Expected episodes

Turn the TVmaze / AniList lookups on or off and set how often airing series are re-checked.

On by default: TVmaze for TV, AniList for anime, and airing series re-checked every 14 days. Lookups are free and need no key.

Quality thresholds

Minimum video bitrate per resolution for the Quality report and the Low-bitrate card. The defaults are in [chapter 11](#).

Posters

Posters are on by default, fetched at 300 px wide, with online lookups allowed for titles Plex does not have. **Fetch missing posters**, **Fetch all again** and a count by source sit here. The folder defaults to `posters` next to the database; see [section 5.4](#) for sharing one folder between the desktop app and the Pi.

22.3 Automation

Schedules and **Notifications** have their own chapters: [chapter 21](#) and [chapter 20](#).

Updates

Automatic updates on/off, and the GitHub token needed only while the repository is private.

On by default. The installed Windows app updates itself from GitHub Releases; on the Pi use `sudo medialedger-update` ([chapter 26](#)).

22.4 Connections

Plex

Server, token, path mapping and sync options: see [chapter 23](#). The **Webhook** row belongs to the web server ([section 23.3](#)); on the desktop it explains that webhooks need the always-on server.

CSV export

Output folder and whether to export after every scan.

The chosen sets, zip options and defaults are in [chapter 17](#).

22.5 Renaming

Enables the TV / anime rename tool. Off by default.

Setting	TV and anime	Movies
Enabled	Off	Off. “Allow live renames” switch; dry runs always work
Batch limit	200	200
Layout	In place	Rename in place, or move into “Title (Year)” folders
Title and year from	The parser and your fixes	File name and fixes, or the Plex match
Name parts	Title (on); resolution, codec, Sub/Dub (opt-in)	Source, resolution, HDR, codec, audio, edition (all on); Sub/Dub opt-in

22.6 Backup and data

Database path and size, schema version and row counts, and the buttons **Back up now**, **Open the backups folder**, **Open the data folder** and **Open the log**. The nightly backup and the restore dialog are described in [section 21.4](#) and [section 21.5](#).

22.7 Appearance

Colour theme switches the whole app between a few palettes and applies at once. The choice is remembered in the browser (or the desktop app) you set it in, not on the server, so every device and every person can have their own.

PART 6 • SETTINGS AND CONNECTIONS

23 Plex integration

WHERE	WHO CAN OPEN IT	IT ANSWERS
Settings → Connections → Plex	Admins	How do I link Plex for ratings, play counts and watch history?

MediaLedger talks to your own Plex Media Server over its local HTTP API. It never writes to Plex and needs no Plex account: only the server's address and a token.

23.1 Setup

1. In Plex Web, open any movie or episode, click the `⋮` menu, choose **Get Info**, then **View XML**.
2. A new tab opens showing XML. Copy its whole address from the browser's address bar; it ends in `X-Plex-Token=...`. Do not share it; it grants access to your server.
3. In MediaLedger, Settings → Plex, paste that address into **Paste the XML address**. The token is pulled out into the token field, the server URL is filled in when the address came from your LAN (for Plex on the NAS, `http://192.168.1.50:32400`), and the pasted text is discarded. Press **Test**. It reports the server version and lists the libraries it can see.
4. Press **Sync now**. The first sync derives the path mapping from the first file it recognises, for example `/media` → `\\192.168.1.50\Media_Share`, and shows it for editing.
5. Tick **Sync after every scan** to keep it current.

TIP The sync defaults to every 6 hours as well as after each scan, so the Watched page and play counts stay current on days without a scan. Set the interval to 0 to sync only after scans.

23.2 What a sync stores

Per file	Per show
Plex title and year, rating key, IMDb / TMDb / TVDB ids, your Plex rating, audience rating, play count, last viewed, resume offset, library section	Plex's title and year, ids, your rating, audience rating, episode and watched counts, content rating

These appear on the Ratings tab (Plex, Plex mine, Watched), on the series lists (Watched %, your Plex rating beside your stars), and as the optional truth source for the movie naming engine. The status line under the Sync button reports how many Plex items matched a file and lists examples of files Plex does not have.

23.3 Plex webhook

A sync pulls from Plex on a schedule; a webhook lets Plex push to MediaLedger the moment something happens. Webhooks are a Plex Pass feature and need a server that is always listening, so they work on the Raspberry Pi web server only. The desktop app shows a note instead.

1. On the Pi, Settings → Plex → **Webhook (Plex Pass)**: switch it on. A URL of the form `http://<pi>:8080/api/plex/webhook?key=...` appears with a copy button.
2. In Plex Web: Settings → **Webhooks** → Add webhook, paste the URL, save.
3. Play or rate something in Plex. The last events received are listed under the switch, so you can confirm the link works.

Plex event	What MediaLedger does
Item added to a library	Queues a scan two minutes later, so a batch of new files arrives in one scan rather than one scan per file.
Watched (scrobble)	Updates the play count and last-viewed time of the linked file at once.
Rated	Stores the new rating on the linked file at once.

CAUTION The key in the URL is the credential: anyone who has it can post events. **New key** generates another and invalidates the old one; update the URL in Plex afterwards. LAN-only still applies, so a Plex server on the internet cannot reach the webhook unless you turn that rule off.

23.4 Per-library breakdown

Under the last-sync line a table breaks the sync down **per Plex library**: items, matched, unmatched and why. *No path mapping covers this folder* means that library lives under a path none of your mappings translate: add a mapping for it. *File is not in a scanned root* means Plex has files MediaLedger does not scan: add the root, or ignore it if that library is not meant to be tracked. The first unmatched path is shown as Plex reports it and as MediaLedger translated it, which is usually enough to see the difference.

PART 7

Server and security

System health, accounts and roles, the Raspberry Pi install and networking.

IN THIS PART

- 24 System and Log**.....71
A live health verdict with history, and the application log in the browser.
- 25 Security**.....74
Roles, what is always on, every control on the page, two-factor codes and HTTPS.
- 26 Raspberry Pi server**.....79
Requirements, the guided installer, first run, connecting shares, daily commands, updating and moving your database over.
- 27 Names, HTTPS and reaching it from outside**.....85
What to publish and what never to, your own LAN name, HTTPS, Caddy and Tailscale Funnel.

PART 7 • SERVER AND SECURITY

24 System and Log

WHERE Sidebar → App → System, and Log	WHO CAN OPEN IT Admins	IT ANSWERS Is the machine healthy, and what does the log say?
---	----------------------------------	---

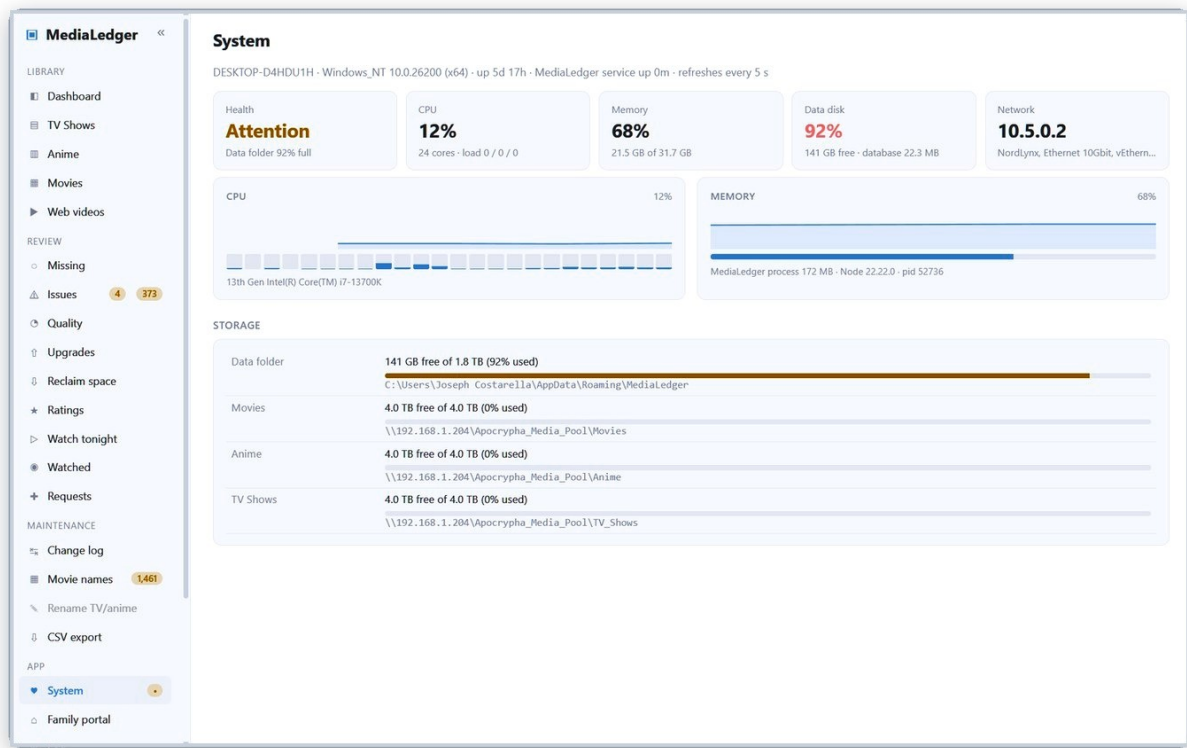


Figure 31. The System page on the desktop: health, CPU per core, memory, data disk, network, and free space on every root.

The System tab watches the machine MediaLedger runs on. On the desktop that is your PC; on the Raspberry Pi it is the board itself, where temperature and power matter. It samples every 5 seconds and keeps one hour of history in memory (nothing is written to the database or the SD card).

Panel	What it shows
Health	One verdict, All good , Attention or Problem , with every triggered rule listed under it. Rules: SoC over 70 °C (attention) or 80 °C (problem); throttling or under-voltage now (problem) or since boot (attention); memory over 85 / 95 %; swapping; a disk over 90 % or under 5 GB free (attention), over 97 % or under 1 GB (problem); a root that cannot be reached while no scan is running. The System entry in the sidebar shows a dot whenever the verdict is not green.
SoC temperature (Pi only)	Current chip temperature, clock speed and core voltage. The Pi firmware soft-limits at 80 °C and throttles at 85 °C; a scan on a caseless Pi 4 usually sits in the 50s to 60s.

Panel	What it shows
Power & throttling (Pi only)	The firmware's throttling flags. Healthy means no under-voltage or frequency capping since boot. "Under-voltage has occurred" means the power supply or cable is weak; fix that before trusting long scans.
CPU	Overall percentage, per-core bars and the 1 / 5 / 15 minute load averages.
Memory	Used against total, swap if any, and the size of the MediaLedger process itself.
Data disk	Free space where the database, log and exports live, and the database size.
Network	Download and upload rate. During a scan the download line is the share being read.
Storage	Free space and fill percentage of the data folder and of every enabled root, measured through the share.

NOTE The thresholds live at the top of `healthOf()` in `src/main/sysmon.js` if you want to tune them for a different case or fan.

24.1 When the verdict changes

The headline tile takes the worst of every rule that triggers. The System entry in the sidebar shows a dot whenever it is not green.

Rule	Attention	Problem
SoC temperature (Pi)	70 °C or more	80 °C or more
Throttling or under-voltage (Pi)	Has occurred since boot: check the power supply	Happening now
Memory used	85 % or more	95 % or more
Swap in use	More than 256 MB	
Load average	Above twice the core count with no scan running	
A disk (data folder or a root)	90 % full or under 5 GB free	97 % full or under 1 GB free
The data folder		Cannot be read
A root		Cannot be reached while no scan is running

24.2 Log

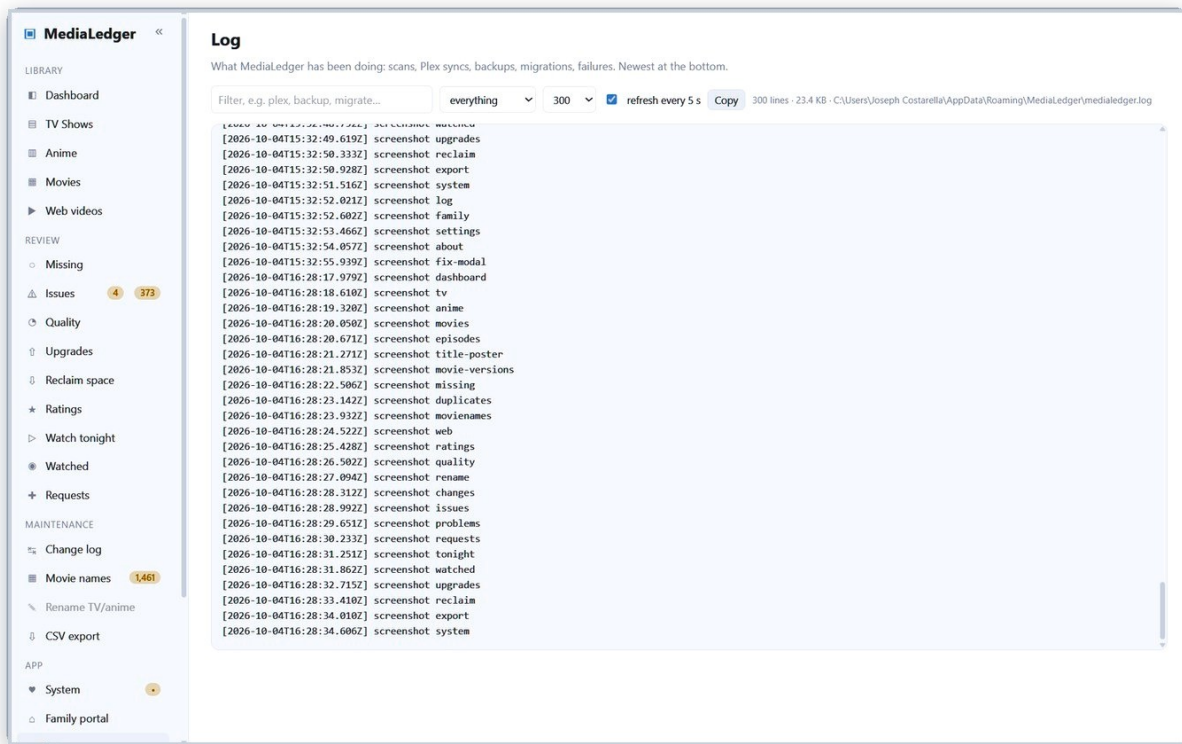


Figure 32. The Log page filtered to Plex lines.

Log shows the tail of `medialedger.log`: scans, Plex syncs and the play-history pull, backups, schema migrations, restores and every failure, newest at the bottom. Type in the filter to narrow it (**plex**, **backup**, **migrate**), switch to **problems only**, load up to 2,000 lines, and copy the result. It refreshes every five seconds while the box is ticked. On the web server only admins see it, so diagnosing the Pi no longer needs SSH.

PART 7 • SERVER AND SECURITY

25 Security

WHERE Sidebar → App → Security (web server)	WHO CAN OPEN IT Admins. Standard users can change their own password	IT ANSWERS Who can get in, what can they do, and how is it protected?
---	--	---

The Security tab manages access to the **web server**. The desktop app has no login: it runs as you, on your PC, and only your PC can reach it, so on the desktop the tab simply says so. On the Raspberry Pi anyone on the LAN could otherwise open the page, and this tab is where you control that.

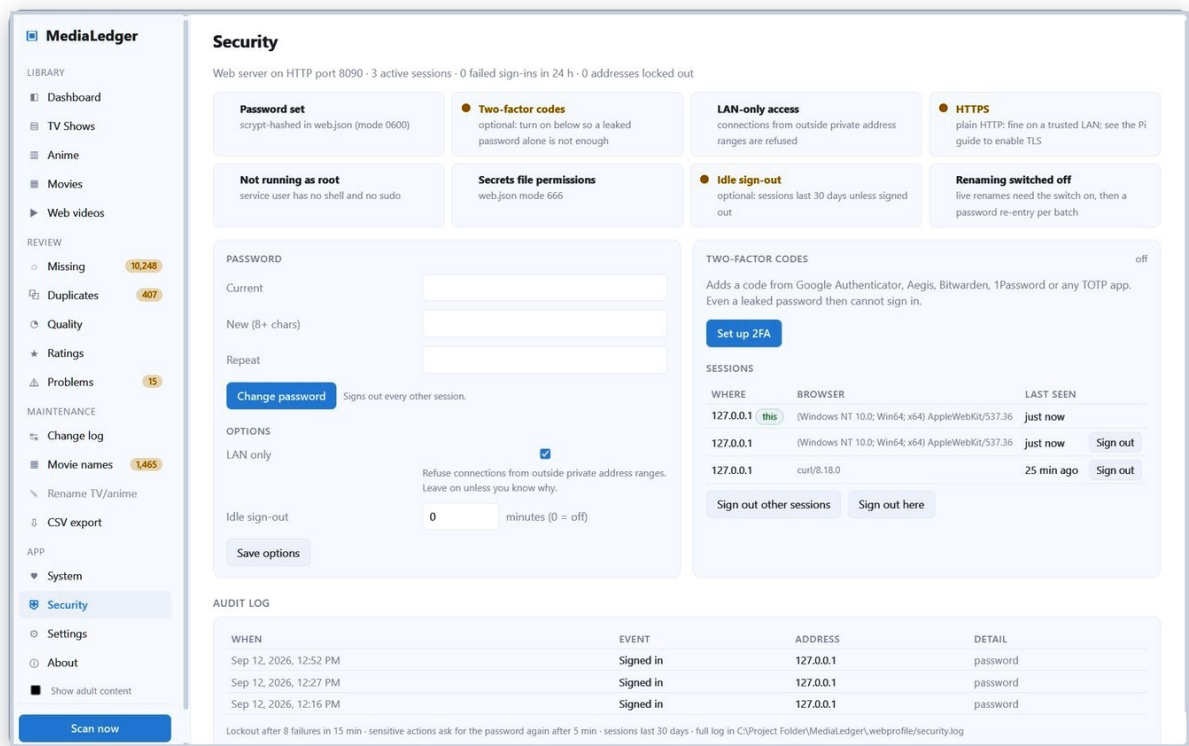


Figure 33. The Security page on the Pi: posture checks, password, options, users, two-factor codes, sessions and the audit log.

25.1 Accounts and roles

Sign-in is a **username and password**. The first account is always **admin**: the installer asks for its password, and an installation upgraded from 1.0 keeps its old password under that name. Admins add further accounts under Security → Users, each with one of three roles:

Role	Can	Cannot
admin	Everything: scans, fixes, renames, settings, security, exports and downloads, request statuses.	

Role	Can	Cannot
standard	See every library and review page, rate titles, file requests, change their own password, and show adult content for their own session.	Settings, System, Security, scans, fixes, renames, exports. The Plex and GitHub tokens are blanked in what they see.
guest	Open the page without signing in (only when Guest access is on): library statistics and lists, and file requests.	Adult content, Issues and every other review page, any control.

Can...	Guest	Standard	Admin
See library lists, charts and statistics	Yes	Yes	Yes
File a media request	Yes	Yes	Yes
Rate titles, write notes, add tags	No	Yes	Yes
Open Review and Maintenance pages (Missing, Issues, Quality, Upgrades, Watched, Reclaim, Change log, Movie names, exports)	No	View	Yes
Show adult content for their own session	No	Yes	Yes
Change their own password	No	Yes	Yes
Run scans, fixes, matches, renames and exports	No	No	Yes
Settings, System, Log, Security, Family portal	No	No	Yes
Change a request's status	No	No	Yes

Every refusal is enforced by the server, not just hidden in the page, and is written to the audit log. Adult visibility is per session: one person switching it on does not show it to anyone else. Two-factor codes, when enabled, apply to admin sign-ins.

25.2 Always on

- Passwords are hashed with scrypt and stored in `web.json` with owner-only permissions. Minimum 8 characters.
- Sessions are HttpOnly, SameSite=Strict cookies that last 30 days. Scripts cannot read them and other sites cannot use them.
- The API accepts JSON only, refuses cross-origin requests, and every page carries a strict Content-Security-Policy plus `X-Frame-Options: DENY, nosniff` and `no-referrer` headers.
- After 8 failed sign-ins in 15 minutes from one address, that address is locked out for 15 minutes.
- **LAN only:** connections from outside private address ranges are refused. Do not port-forward the Pi; use a VPN into your LAN if you need remote access.
- **Re-authentication:** a live movie rename, an undo, a TV/anime rename, purging missing files and every change on this tab ask for the password again unless you entered it within the last 5 minutes. A small dialog appears in place; the action continues once you confirm.
- Every sign-in, failure, logout, re-authentication, sensitive action and setting change is written to `security.log` and listed at the bottom of the tab.

Limit	Value
Minimum password length	8 characters
Web session	30 days, or the idle sign-out you set
Failed sign-ins before logout	8 in 15 minutes from one address
Lockout	15 minutes
Password re-entry for dangerous actions	Not asked again for 5 minutes after you enter it
Family portal: remembered device	180 days
Family portal: admin session	12 hours, or 1 hour idle
Family portal: requests per person	6 an hour, 25 a day

25.3 Controls

Control	What it does
Posture checks	Coloured tiles at the top: admin account, two-factor, LAN-only, guest access, HTTPS, not running as root, secrets file permissions, share credentials permissions, idle sign-out, renaming switched off. Green is good; amber is an optional improvement; red needs attention.
Change password	Changes the password of the signed-in account; needs the current one. Signs out every other session. Standard users get this control alone.
LAN only	On by default. Turn off only if you know exactly which non-private network should reach the Pi.
Idle sign-out	Ends a session after N minutes without activity. 0 keeps the 30-day limit only. The Save beside the field and Save options below do the same thing.
Guest access	Lets anyone on the LAN open the page without signing in, as a guest (section 25.1). Off by default.
Users	Add an account (name, password, role), change a role, reset a password, delete an account. Admins only; every change asks for your password again.
Two-factor codes	Press Set up 2FA , scan the QR code with an authenticator app and enter the 6-digit code it shows (section 25.4). From then on admin sign-in needs password + code. Turning it off needs the password. If you lose the device, see the Pi troubleshooting table.
HTTPS	Creates a self-signed certificate and restarts the server on it (section 25.5). Once on, a Download certificate button and per-device instructions replace the switch.
Guest link	Shown once Guest access is on: a QR code of the site address that anyone on the Wi-Fi can scan to open MediaLedger as a guest.
Sessions	Every signed-in browser with its address, browser and last activity. Sign out ends one; Sign out other sessions keeps only this one; Sign out here ends yours.
Audit log	The last 100 events with time, kind, address and detail. The full log is <code>security.log</code> in the data folder.

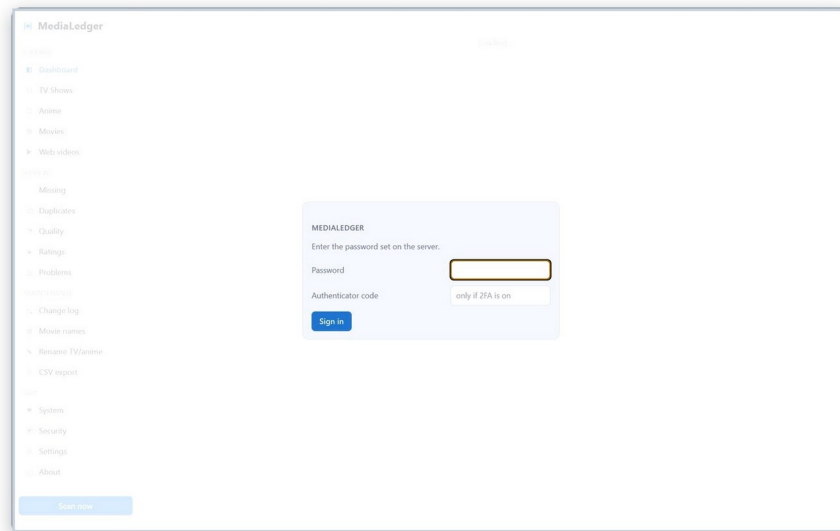


Figure 34. Signing in to the web server with a username and password. The code field is only needed once two-factor is on.

25.4 Two-factor codes with a phone

1. Install **Google Authenticator** (or Aegis, Bitwarden, 1Password, Microsoft Authenticator: any app that does time-based codes) on your phone.
2. On the Pi, Security → Two-factor codes → **Set up 2FA**. A QR code appears.
3. In the app tap + → **Scan a QR code** and point the camera at the screen. The app adds an entry named MediaLedger and starts showing a 6-digit code that changes every 30 seconds. No camera? Tap **Enter a setup key** and type the secret shown under the code.
4. Type the current 6-digit code into the box under the QR and press **Turn on 2FA**. The server checks it before switching on, so a mistyped secret cannot lock you out.
5. From now on the sign-in dialog asks for the code after the password, for admin accounts. Standard users and guests are unaffected.

NOTE The secret lives in `web.json` on the Pi and in the app on your phone; nothing is sent to Google or anyone else. The Pi clock must be roughly right (`timedatectl`), since the codes are computed from the time.

25.5 HTTPS

When a reverse proxy on the same machine (Caddy, see [section 27.3](#)) serves HTTPS in front of MediaLedger, the server notices the first time you open the site through it: the HTTPS line in the checklist turns green, the heading reads **on, by the reverse proxy**, and the button for the built-in certificate is not offered, because turning it on would break the proxy. Only a proxy on the machine itself is believed; another computer on the network cannot claim it.

On a home LAN plain HTTP is normal, and MediaLedger refuses connections from outside the LAN anyway. HTTPS adds encryption between each browser and the Pi, which matters if you do not fully trust every device on the Wi-Fi, and it lets phones add the site as a proper app without warnings. There is no certificate authority for a private LAN name, so MediaLedger makes its own **self-signed** certificate; browsers trust it once you install it on each device.

1. Security → HTTPS → **Turn on HTTPS**, then confirm. The server writes a certificate valid for ten years covering every name it answers to (`medialedger.local`, the hostname, your custom domain if the installer was given one, and every LAN address), audits the change, and restarts. Sign-in sessions survive the restart.
2. The page reopens on `https://`. If the Pi was installed with `--port=80`, the site moves to port 443 (no port in the address) and port 80 redirects there, so `http://medialedger.home` keeps working.
3. The browser warns that the certificate is not trusted. Either accept the warning for this site, or remove it for good: press **Download certificate** (a `.crt` file Windows opens with its own installer on double-click) and follow the per-device steps under **Removing the browser warning** (Windows: Trusted Root Certification Authorities; Android: install a CA certificate; iPhone: install the profile, then Certificate Trust Settings; macOS: Keychain, Always Trust).

CAUTION Adding a new name for the Pi later (a custom domain, a rename) needs a new certificate: delete `/var/lib/medialedger/tls/` on the Pi, restart the service, and turn HTTPS on again. Installing the certificate on a device is per device; the file itself is public, only `key.pem` on the Pi is secret.

PART 7 • SERVER AND SECURITY

26 Raspberry Pi server

WHERE The Pi, then any browser on the LAN	WHO CAN OPEN IT You, as admin	IT ANSWERS How do I install, run, update and move to the Pi?
---	---	--

The same program runs as a small always-on website on a Raspberry Pi. Every page in this manual works there, in any browser on your LAN. The Pi keeps its own database and settings. The format is identical to the desktop, so a database can be copied between the two.

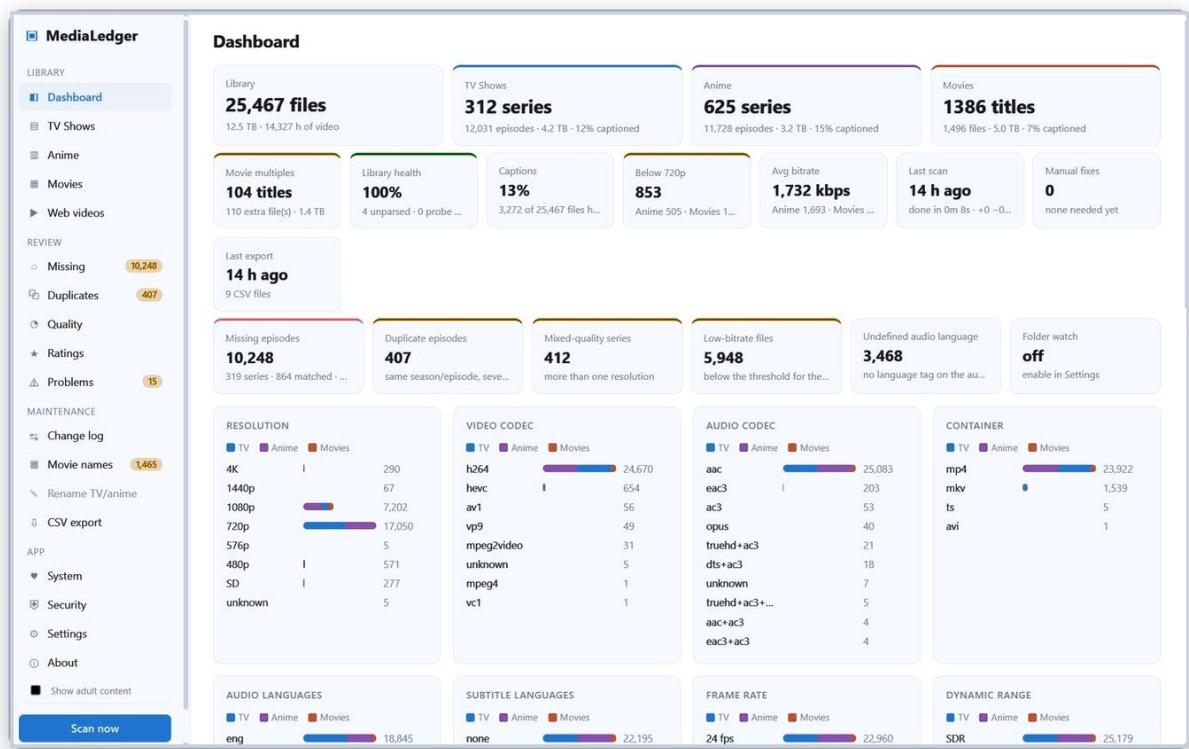


Figure 35. The dashboard served from the Pi, in a browser.

	Windows desktop	Raspberry Pi
Choosing folders	Browse, or type a UNC path	The picker lists the connected shares. Typing a path also works
“Open folder” and “Reveal” buttons	Open Explorer	Do nothing in a browser
Updates	Silent installer, checked every six hours	<code>sudo medialedger -update</code> , or a nightly timer
Scheduling a scan	In-app timer, or Windows Task Scheduler	The in-app timer. The Task Scheduler section does nothing
Sign-in and the Security page	None. Only your PC can reach it	Accounts, roles, optional two-factor

	Windows desktop	Raspberry Pi
Plex webhook and Home Assistant status	Not available	Available

26.1 Requirements

Item	Minimum	Recommended
Board	Raspberry Pi 3 (64-bit OS)	Raspberry Pi 4B or 5 , any RAM size (2 GB is plenty)
Storage	16 GB microSD, A1 class	32 GB A2 microSD or a USB SSD. A cheap card wears under log and database writes
Network	Wi-Fi	Ethernet. A Pi 4 scans at near-PC speed; a Pi 3 (100 Mbit) takes hours for the first full scan
Power	Official supply	Same. Under-voltage shows on the System page
Operating system	Raspberry Pi OS Lite (64-bit), Trixie or newer	Same. 64-bit is required: Node 22's built-in SQLite only has arm64 builds
Installed by the script	Node 22, ffmpeg, cifs-utils, smbclient, curl, openssl	
You need to know	The file server's address, a username and password that may read the media, and a web password of 8 or more characters	

26.2 The install in six steps

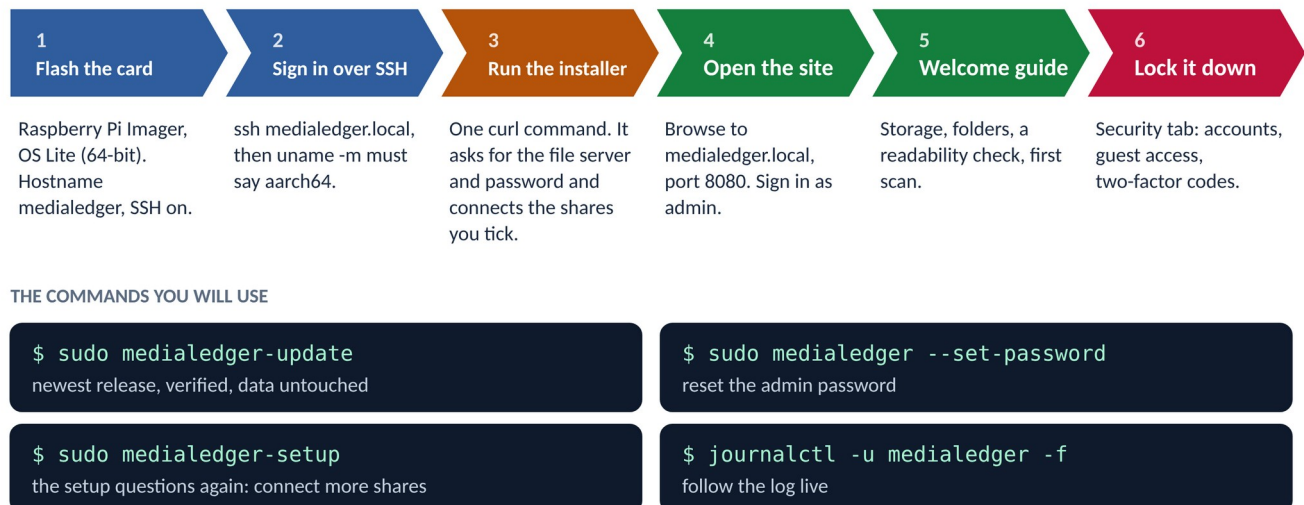


Figure 36. From an empty card to a locked-down service, and the four commands you will use afterwards.

26.3 Install, step by step

1. Flash **Raspberry Pi OS Lite (64-bit)** with Raspberry Pi Imager: Choose OS → Raspberry Pi OS (other) → Raspberry Pi OS Lite (64-bit). In the customisation step set hostname `medialedger`, your user and password and time zone, and enable SSH.
2. Boot the Pi on Ethernet. From your PC run `ssh medialedger.local`, then `uname -m`. It must print `aarch64`.
3. Run the installer on the Pi. It asks its questions on the terminal even though the script arrives on a pipe:

ON THE PI

```
curl -fsSL https://raw.githubusercontent.com/AxialForge/medialedger/main/server/install.sh -o install.sh && sudo bash install.sh
```

The installer is safe to rerun. It:

1. installs Node 22, ffmpeg, cifs-utils, smbclient, curl and openssl;
2. creates the `medialedger` system user (no shell, member of `video` for the hardware readings) and `/var/lib/medialedger`;
3. downloads the **server-only release package** from GitHub Releases, verifies its SHA-256 and unpacks it to `/opt/medialedger`;
4. asks for the **file server's address, username and password**, lists the shares it offers, and mounts the ones you pick under `/mnt/medialedger/<share>`. Press Enter at the address question to skip and connect shares later from the web app;
5. installs a hardened systemd service on port 8080, a one-minute mount watchdog, the share helper, a status banner for SSH sign-ins and the commands `medialedger`, `medialedger-update` and `medialedger-setup`;
6. asks for the **admin password** (8 or more characters) if none is set, then prints the address to open.

Installer option	Effect
<code>--setup</code>	Only the questions: connect shares and set the password. Also <code>sudo medialedger-setup</code>
<code>--share=//host/share</code>	Mount one share at <code>/mnt/media</code> without being asked, as installs before 2.4 did
<code>--port=9000</code>	Serve on another port. <code>--port=80</code> works: the unprivileged service is granted only the capability to bind low ports
<code>--domain=medialedger.home</code>	Remember a LAN name for the Pi. The HTTPS certificate is issued for it (section 27.1)
<code>--https</code>	Also create a self-signed certificate and serve HTTPS
<code>--update-only</code>	Used by <code>medialedger-update</code> . Also moves the service to another port without a reinstall
<code>--auto-update</code> / <code>--no-auto-update</code>	Add or remove a systemd timer that runs the update every night at about 04:30
<code>--branch=main</code>	Run from a git branch instead of the release package (development)

26.4 First run on the Pi

1. Open `http://medialedger.local:8080` (or `http://<pi-address>:8080`) and sign in as `admin` with the password you gave the installer.
2. The **welcome guide** opens by itself on a new install: connect storage, pick the TV, Anime and Movies folders (**Find them for me** looks inside the connected shares for folders with the usual names), check that every folder can be read, and start the first scan. Reopen it any time from Settings → Library.
3. Add the Web videos and Adult roots under Settings if you use them.
4. The first scan probes every file, so you can close the browser and come back. Or copy the desktop database over first (section 26.8) and skip it.
5. Open **System** during the scan to see temperature and network load, and **Security** afterwards to add accounts, decide on guest access and set up two-factor codes.

26.5 Connecting more shares

Connect a share from Settings → Library → **Network shares**, from the folder picker, or with `sudo medialedger-setup`. Each appears under `/mnt/medialedger/<name>` and as a button at the top of the folder picker, which lists up to 2,000 folders, sorts “Season 2” before “Season 10”, and says how many files a folder holds when it has no sub-folders.

The web service itself has no root and cannot mount anything. A **share helper** started as root by systemd does it. It treats every request as hostile: it checks the owner and size, matches every field against a pattern, and mounts only under `/mnt/medialedger/`. The sign-in goes in a root-only file in `/etc/medialedger-shares/`. If a mount fails the helper puts fstab back and deletes the credentials, because a stale `cifs` line can hang a boot. An install from before 2.4 keeps its single share at `/mnt/media` unchanged.

26.6 Daily commands

Command (on the Pi)	Does
<code>sudo medialedger-update</code>	Download the newest release, verify it, install it and restart. Data is untouched
<code>sudo medialedger-setup</code>	The guided setup again: connect more shares, set the password, show the addresses
<code>sudo medialedger --set-password</code>	Set or reset the admin password. Signs everyone out
<code>systemctl status medialedger</code>	Running? Since when? The last log lines
<code>journalctl -u medialedger -f</code>	Follow the application log live
<code>journalctl -u medialedger -n 50 --no-pager</code>	The last 50 log lines
<code>sudo systemctl restart medialedger</code>	Restart the service (also stop and start)
<code>ls /mnt/medialedger</code>	The connected shares, one folder each
<code>sudo mount /mnt/medialedger/<name></code>	Re-mount by hand after a NAS reboot. The watchdog also does it every minute

Command (on the Pi)	Does
<code>sudo bash install.sh</code>	Rerun the installer: repairs the service, mounts and commands, keeps data
<code>vcgencmd measure_temp</code>	Chip temperature from the shell. The System page shows the same

NOTE To change a share's sign-in, remove it under Settings → Library → Network shares and connect it again. On an install from before 2.4, delete `/etc/medialedger-cifs.cred` and rerun the installer.

26.7 Updating

`sudo medialedger-update` downloads the newest release package, verifies it, swaps it into `/opt/medialedger` and restarts the service. Your database and settings live in `/var/lib/medialedger` and are never touched. Schema upgrades run on start with a backup taken first, as on the desktop. Rerun the installer once with `--auto-update` to add a nightly timer; without it admins see a notice on the Dashboard when a newer release exists.

26.8 Moving the desktop database to the Pi

This carries every fix, keep decision, series match, rating and the whole change log across, and skips the first full scan. On the PC, close MediaLedger and copy the database to the Pi:

ON THE PC

```
scp "%APPDATA%\MediaLedger\medialedger.db" you@medialedger.local:/tmp/medialedger.db
```

On the Pi:

ON THE PI

```
sudo systemctl stop medialedger && sudo mv /tmp/medialedger.db
/var/lib/medialedger/medialedger.db && sudo chown medialedger:medialedger
/var/lib/medialedger/medialedger.db && sudo systemctl start medialedger
```

Then set the roots in Settings with the same ids as on the desktop (movies, anime, tv and any you added) and the `/mnt/media/...` paths, and run a scan. Every file is recognised by its relative path and nothing is re-probed. Do not copy `settings.json`: it holds Windows paths.

26.9 File locations

Path on the Pi	Contents
<code>/opt/medialedger</code>	The application (replaced on update)
<code>/var/lib/medialedger/medialedger.db</code>	The database
<code>/var/lib/medialedger/settings.json</code>	Settings (roots, schedule, Plex, ...)
<code>/var/lib/medialedger/web.json</code>	Accounts and password hashes, sessions, 2FA, options, webhook key (owner-only)
<code>/var/lib/medialedger/security.log</code>	Audit log

Path on the Pi	Contents
<code>/var/lib/medialedger/medialedger.log</code>	Application log
<code>/var/lib/medialedger/exports/</code>	CSV exports
<code>/var/lib/medialedger/backups/</code>	Pre-migration database backups
<code>/var/lib/medialedger/tls/</code>	Optional <code>cert.pem</code> and <code>key.pem</code>
<code>/mnt/medialedger/<name></code>	Each connected share
<code>/etc/medialedger-shares/<name>.cred</code>	Share sign-ins (root-only). Older installs: <code>/etc/medialedger-cifs.cred</code>
<code>/etc/medialedger-domain</code>	The LAN name given to the installer
<code>/etc/systemd/system/medialedger.service</code>	The service definition

The service runs as the `medialedger` user with no shell, `NoNewPrivileges`, `ProtectSystem=full`, `ProtectHome` and a private `/tmp`. It can write only to its data folder and the shares. Account hashes, sessions, the 2FA secret and the webhook key are owner-only in `web.json`.

PART 7 • SERVER AND SECURITY

27 Names, HTTPS and reaching it from outside

WHERE	WHO CAN OPEN IT	IT ANSWERS
The Pi, your router and Tailscale	You, as admin	How do I give it a name, add HTTPS and share the portal safely?

Out of the box the Pi answers at `http://medialedger.local:8080` on your LAN and refuses everything else. This chapter changes that in small, safe steps. The rule behind all of them: the **web app stays on the LAN**, and only the **family portal** is ever published.

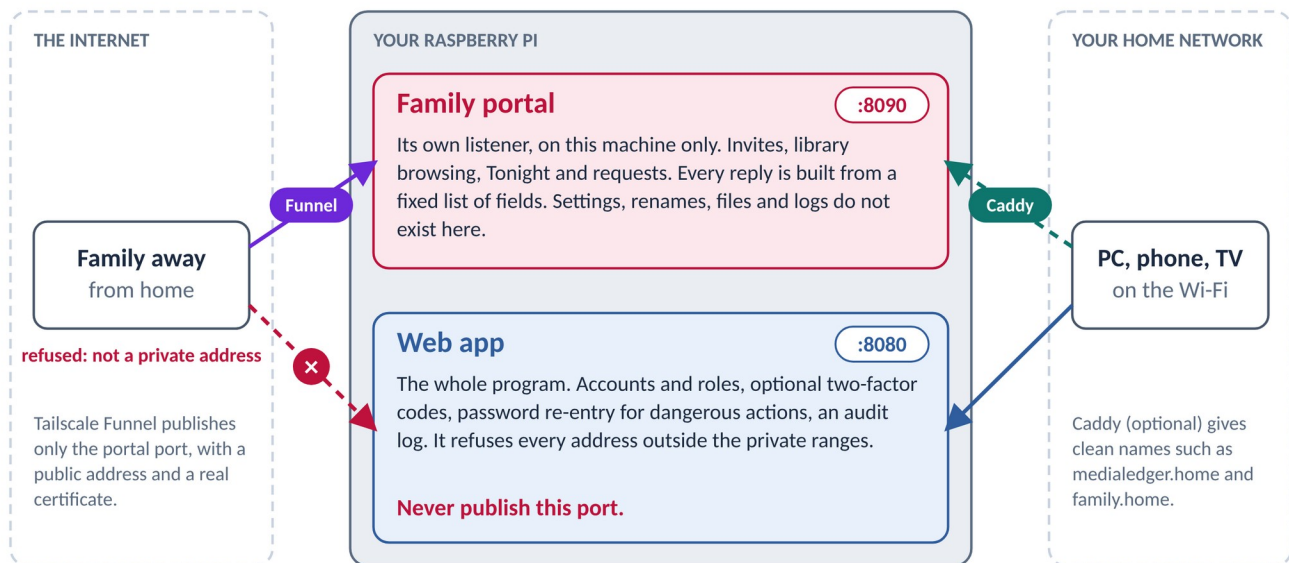


Figure 37. What can reach what. The web app refuses public addresses. The portal is a separate listener meant to sit behind a proxy.

- Never port-forward the Pi. For admin access from outside, use a VPN into your LAN (WireGuard or Tailscale); the LAN-only rule keeps working.
- To let visitors in, publish only the portal port, 8090, with Tailscale Funnel ([section 27.4](#)).
- A name in your router, or in Caddy, is invisible to the internet. None of the steps below exposes the Pi.

27.1 Your own name: medialedger.home

`.local` names come from mDNS, which Windows, Mac, iPhone and Android resolve without setup, so `medialedger.local` already works as an internal name. Two things make it feel like a real site: dropping the `:8080` and choosing your own name.

Pick a name that no public registry owns and that your router can resolve: `medialedger.home`, `medialedger.lan`, or the reserved `medialedger.home.arpa`. Avoid `.local` for this (mDNS reserves it and it breaks if a DNS server also answers it) and avoid real public suffixes such as `.com` unless you own the domain.

1. Tell your router the name

On a UniFi network (Network 8 or newer): make the Pi's lease fixed (Client Devices → medialedger → Settings → Fixed IP Address), then Settings → Routing → **DNS** → Create Entry: type **A**, domain name `medialedger.home`, the Pi's address. On a client, `nslookup medialedger.home` should answer within a minute. Devices with a hard-coded public DNS such as 8.8.8.8 will not see it.

Other routers call the feature **Local DNS**, **DNS records**, **Static hostnames** or **Host overrides** (pfSense and OPNsense). Pi-hole and AdGuard Home have a **Local DNS records** page. With none of those, a line in each device's hosts file does the same for that device:

HOSTS FILE

```
192.168.1.50 medialedger.home
```

2. Serve on the default port

Rerun the installer with the port and the name. The service keeps running as the unprivileged `medialedger` user; the unit grants only the one capability needed to bind port 80. The name is remembered in `/etc/medialedger-domain` for later reruns.

ON THE PI

```
sudo bash install.sh --port=80 --domain=medialedger.home
```

`http://medialedger.home` now opens the app.

27.2 HTTPS for the web app

The easy way is the **Turn on HTTPS** button on the Security page (section 25.5). From the shell, add `--https` once and the self-signed certificate is issued for `medialedger.home`, `medialedger.local` and the IP, on the port you chose. With `--port=80` the site moves to 443, so `https://medialedger.home` needs no port, and port 80 redirects there. A public certificate (Let's Encrypt) needs a domain you own and is not needed on a LAN.

The same certificate by hand:

ON THE PI

```
sudo mkdir -p /var/lib/medialedger/tls && sudo openssl req -x509 -newkey rsa:2048 -nodes -days 3650 -subj "/CN=medialedger.local" -keyout /var/lib/medialedger/tls/key.pem -out /var/lib/medialedger/tls/cert.pem && sudo chown -R medialedger:medialedger /var/lib/medialedger/tls && sudo chmod 600 /var/lib/medialedger/tls/*.pem && sudo systemctl restart medialedger
```

Device	To stop the browser warning, install the certificate here
Windows	Trusted Root Certification Authorities (double-click the downloaded <code>.crt</code>)
macOS	Keychain Access, then set it to Always Trust
Android	Settings → install a CA certificate
iPhone	Install the profile, then Certificate Trust Settings

The Security page offers **Download certificate** with these steps once HTTPS is on. The certificate file is public; only `key.pem` on the Pi is secret.

27.3 Several services on one Pi: Caddy in front

DNS records carry an address, never a port, so a second web service on the same Pi either lives on its own port in the URL or sits behind a reverse proxy. **Caddy** is the easy proxy: it owns ports 80 and 443, reads the name the browser asked for, and hands the request to the right service on a local port. Every service then has a clean `https://name.home` address with no port, and one root certificate covers all of them.

Setup, once:

ON THE PI

```
sudo apt install caddy
sudo bash install.sh --update-only --port=8080
```

Turn MediaLedger's own HTTPS off (Security → **Turn HTTPS off**), then write one block per service in `/etc/caddy/Caddyfile` and reload:

/ETC/CADDY/CADDYFILE

```
medialedger.home {
    tls internal
    reverse_proxy 127.0.0.1:8080
}

family.home {
    tls internal
    reverse_proxy 127.0.0.1:8090
}
```

ON THE PI

```
sudo systemctl reload caddy
```

`tls internal` makes Caddy issue certificates from its own authority. Install its root once per device from `/var/lib/caddy/.local/share/caddy/pki/authorities/local/root.crt` and no service on that Pi warns again. To let visitors in over plain HTTP without a certificate, name both schemes: `http://medialedger.home, https://medialedger.home { ... }`.

Adding a service later is one block in the Caddyfile on the next free port, a reload, and one DNS record for the new name pointing at the same Pi. Behind the proxy MediaLedger sees every visitor as 127.0.0.1 unless the proxy forwards the real address, which Caddy does by default.

27.4 The portal from outside: Tailscale Funnel

People away from your home need a public address for the portal. **Tailscale Funnel** gives one for free, with a real certificate, without buying a domain and without opening a port on the router. Only the portal port is published; the admin site and every other service on the Pi stay private.

1. Create a free Tailscale account at tailscale.com (a Google, Microsoft, GitHub or Apple sign-in works).
2. On the Pi, install it and bring it up. It prints a link; open it on your PC and approve the Pi:

ON THE PI

```
curl -fsSL https://tailscale.com/install.sh | sh
sudo tailscale up
```

1. In the Tailscale admin console, under DNS, enable **MagicDNS** and **HTTPS certificates**. Under Access controls add the **funnel** node attribute when the console offers it the first time you run the next command.
2. In MediaLedger open **Family portal**, tick **on**, keep port 8090 and **this machine only**, and Save.
3. On the Pi publish the portal port. It prints the public address, like `https://aether.your-tailnet.ts.net`:

ON THE PI

```
sudo tailscale funnel --bg 8090
```

1. Paste that address into **Public address** on the Family portal page and Save. Create an invite and open its link on a phone that is on mobile data.

`tailscale funnel status` shows what is published, and `sudo tailscale funnel --bg --https=443 off` stops it. Funnel survives reboots. At home you can also serve the portal under its own name with the `family.home` block above and a DNS record for the name.

IMPORTANT Funnel publishes only what you point it at. Never point it at the main MediaLedger port.

27.5 Before you publish anything

- Two-factor codes are on for admins (section 25.4), especially if you ticked **Admin sign-in from outside**.
- The portal is set to **this machine only**, so only Funnel or Caddy can reach it.
- Funnel points at 8090 and nothing else. `tailscale funnel status` confirms it.
- From a phone on mobile data the portal link works and the main app's address does not.
- Guest access is off unless you want anyone on the LAN to browse without an account.

PART 8

Reference

Troubleshooting, defaults and limits at a glance, and the glossary.

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	Every term used in this manual.	

PART 8 • REFERENCE

28 Troubleshooting

WHERE	WHO CAN OPEN IT	IT ANSWERS
Everywhere	Everyone	Something is wrong. What do I check first?

Start with the log. On Windows it is `%APPDATA%\MediaLedger\medialedger.log`; Settings → Data → **Open log** opens it. On the web server admins see it on the **Log** page (section 24.2), and on the Pi `journalctl -u medialedger -n 50 --no-pager` shows the same lines.

28.1 Scanning and the library

Symptom	Cause and fix
Scan finishes instantly and the change log says root_offline	The path in Settings is wrong or the NAS is off. Fix the path. The files are untouched and never marked missing.
Files show but resolution and length are empty	ffprobe was not found when they were scanned. Check Settings → ffprobe, download it if needed, then scan again. Unprobed files are probed automatically.
A series is missing episodes it clearly has	Either the parser could not place some files (Issues → Problems → Unparsed, use Fix...) or the online match is wrong (use Match... and pick the right entry or enter counts).
Missing shows “absolute numbering”	Episode numbers on disk continue across seasons. That season is skipped. Enter per-season counts by hand, or fix the episode numbers with Fix...
A series shows “no match”	The online lookup could not find it. Use Match... to search TVmaze or AniList by hand, or enter the counts yourself.
The Adult page is not there	Either no Adult root is enabled, or the Show adult content switch is off. It resets on every launch.
The app will not start a second copy	Only one instance runs. Launching again focuses the open window, and a scheduled <code>--scan</code> hands its request to it.
The first scan is slow	Probing, not listing, sets the pace. Use Ethernet, not Wi-Fi, and watch System → Network. Over gigabit expect about 25 minutes for 24,000 files.

28.2 Renaming

Symptom	Cause and fix
A movie name is blocked as a collision	Two files would get the same name. Keep one and ignore the other under Name collisions , or give one an edition with Fix...
A rename batch aborted in pre-flight	Something changed since the last scan: a size differs, the target exists or the share is read-only. The report names each problem. Fix or deselect those files and run again. Nothing was renamed.

Symptom	Cause and fix
Renames fail: root not writable	On the Pi the mount is owned by <code>medialedger</code> with write modes. Check that the NAS user has write permission on the share.
A file already has (Year) or Source in its name	It was renamed with a placeholder. Plex will not match it. Fill the value with Fix... and rename again (section 15.7).

28.3 Plex

Symptom	Cause and fix
A sync matched 0 items	The path mapping is wrong. Compare a Plex file path (from View XML) with the same file's path in MediaLedger and set the mapping under Settings → Plex.
Unmatched items in one library	Read the per-library table (section 23.4): <i>*No path mapping covers this folder*</i> needs a mapping, <i>*File is not in a scanned root*</i> needs a root or an ignore.
The webhook shows no events	The URL in Plex Web → Settings → Webhooks must match the one in Settings → Plex, Plex Pass must be active, and the Plex server must be on the LAN (LAN-only refuses others).

28.4 The web server and the Pi

Symptom	Check
The page does not load	<code>systemctl status medialedger</code> and <code>journalctl -u medialedger -n 50</code> . Is the PC on the same LAN? LAN-only refuses other address ranges.
Sign-in loops	Cookies are blocked for the site, or the Pi clock is far off (<code>timedatectl</code>).
"Too many failed attempts"	A 15-minute lockout for that address. Wait, or restart the service.
Roots show "not reachable"	<code>ls /mnt/medialedger</code> . If a share is empty try <code>sudo mount /mnt/medialedger/<name></code> . Wrong credentials: remove the share under Settings → Library → Network shares and connect it again.
A person sees no Settings, System, Log, Security or Family portal	They have the standard or guest role. Only admins see those pages. Change the role under Security → Users.
Forgot the admin password	<code>sudo medialedger --set-password</code> . An admin resets a standard user's password under Security → Users.
Lost the 2FA device	Stop the service, set <code>"totp": {"enabled": false}</code> in <code>/var/lib/medialedger/web.json</code> , start it again.
The Power & throttling tile is missing	The service user must be in the video group: <code>sudo usermod -aG video medialedger</code> && <code>sudo systemctl restart medialedger</code> . The installer does this.
Above 80 °C during scans	Add a heatsink or a fan, or lower Parallel fprobe processes in Settings.
The HTTPS warning will not go away	Install the certificate on that device (section 27.2). A new name for the Pi needs a new certificate: delete <code>/var/lib/medialedger/tls/</code> , restart, and turn HTTPS on again.

28.5 The family portal

Symptom	Cause and fix
The address check turns amber or red	It tests the name the way a visitor would: DNS, connection, certificate and the portal's own answer. The message on the page says which step failed.
"The address refused the secure connection"	The proxy in front has no certificate ready for the name. With Tailscale run <code>tailscale funnel status</code> and check that HTTPS certificates are on for the tailnet.
An invite link says it no longer works	It expired, was revoked, or was replaced by New link . Create or renew the invite.

PART 8 • REFERENCE

29 Quick reference

WHERE Reference	WHO CAN OPEN IT Everyone	IT ANSWERS What is the default, the limit, the command or the path?
---------------------------	------------------------------------	---

29.1 Defaults at a glance

Setting	Default
Roots	Movies, Anime and TV Shows, all enabled
Parallel ffmpeg processes	8
Scan threads	Automatic: CPU count – 1, minimum 2, maximum 32
Re-probe unchanged files	Off
Folder watch	Off. Settle time 90 seconds
Root reachability check	Every 5 minutes
In-app scan timer	Off. 24 hours when on
Windows scheduled task	Off. Daily at 03:00, named “MediaLedger Scan”
Expected episodes	On. TVmaze for TV, AniList for anime, airing series re-checked every 14 days
Plex	Off. When on, sync after every scan and every 6 hours
Export after every scan	On. Sets: TV, Anime, Movies, Web videos, Change log. Zip off, minimum 4 files
Posters	On. 300 px wide. Online lookups on
Daily snapshot	03:05
Daily summary	08:00
Nightly backup	Off. 03:30, newest 7 kept
Renaming (TV/anime and movies)	Off. Batch limit 200
Movie layout and truth source	Rename in place. Title and year from the file name and your fixes
Select next	10 files
Reclaim space	12 months, 2 GB minimum, 4 ★ titles protected
Next up	Last 60 days
Notifications	No channel. All six events on
Auto-update (Windows)	On. Checked at launch and every six hours
Adult content	Hidden on every launch. Excluded from CSVs

Setting	Default
Theme	Dark

29.2 Ports and web addresses

Address	What it is	Who can use it
:8080	The web app (the Pi default)	Private addresses only
:8090	The family portal	Whoever the proxy or Funnel lets through
/request	The phone-sized request form	Anyone allowed in, including guests when guest access is on
/api/status?key=...	Read-only JSON for Home Assistant (section 20.3)	Anyone with the key, on the LAN
/api/plex/webhook?key=...	Where Plex posts events (section 23.3)	Anyone with the key, on the LAN
/poster/<type>/<key>	Poster images	Signed-in users, and guests when guest access is on
/exports/...	CSV and zip downloads	Admin sessions
/tls/medialedger-cert.crt	The public half of the self-signed certificate	Any signed-in session, when HTTPS is on
<public address>/admin	Admin sign-in on the portal (section 19.5)	Admins, when the option is on

29.3 Command lines

Where	Command	Does
Windows	<code>MediaLedger.exe --scan</code>	Scan, export and quit. If the app is open, the open window runs the scan
Pi	<code>sudo medialedger-update</code>	Install the newest release
Pi	<code>sudo medialedger-setup</code>	Connect shares, set the password
Pi	<code>sudo medialedger --set-password</code>	Reset the admin password
Pi	<code>sudo bash install.sh [options]</code>	Install or repair. Options are in section 26.3
Pi	<code>journalctl -u medialedger -f</code>	Follow the log

The full list of Pi commands is in [section 26.6](#).

29.4 File and folder names the parser understands

Anything the parser cannot place is still indexed and probed. It appears under **Issues → Problems → Unparsed file names** with a **Fix...** button. When the parser rules improve in a new version, the whole database is re-parsed on the next launch, so you get the improvement without a rescan.

Layout	Parsed as
Show\Show S1\Show S01E01 - 720p WEB-DL.mp4	S01E01. The release tag is not taken as a title
Show\Show S1\12 Show S1.mp4 and 2Show S1.mp4	Episode 12 and episode 2. The season comes from the folder or the S1 suffix
Show\Show_S01E03_Episode Title_.mp4	S01E03 with an episode title
Show\Group_Romaji_Title_-_01_720p_Group.mp4	Episode 1
Show\Season 02\Show - 2x05 - Title.mkv	S02E05
Show S01E01E02.mkv and S01E01-02	A double episode
Show\Show S0\..., Specials, Show - OVA.mp4, Special 01 - Title.mp4	Season 0
Title (Year) [720p].mp4 and Title (Year) [4k].mkv	One title, two versions
Title (Year)\Title (Year) - Extended.mkv	The edition comes from the file name

The show name comes from the **top** folder, so a typo in a season folder name does not change it. Files directly under a show folder default to season 1.

PART 8 • REFERENCE

30 Glossary

WHERE Reference	WHO CAN OPEN IT Everyone	IT ANSWERS What does that word mean?
---------------------------	------------------------------------	--

Word	Meaning
ffprobe	Part of ffmpeg; reads a media file's streams and reports resolution, codecs, languages and more without playing it.
Sidecar	A subtitle, poster or metadata file that sits next to a video with the same base name.
Override / fix	A manual correction stored in the ledger and re-applied on every scan.
Placeholder	A literal word written into a movie name where the engine could not prove a value.
Pre-flight	The set of checks run over an entire rename batch before any file is touched.
Journal	The per-file record of a rename batch that makes undo possible.
Truth source	Where the naming engine takes title and year from: file name plus fixes, or the Plex match.
Root	A watched folder with a library type.
Rating key	Plex's internal id for an item.
TOTP	Time-based one-time password: the 6-digit codes an authenticator app shows, used for two-factor sign-in on the web server.
LAN only	The web server rule that refuses connections from public address ranges.
Re-authentication	Entering the password again before an action that changes files or security settings.
Role	What an account on the web server may do: admin, standard or guest.
Guest	Someone using the web server without signing in, allowed only when Guest access is on.
Webhook	A URL Plex calls when something happens on the Plex server, so MediaLedger reacts at once instead of on the next sync.
Card	A tile, chart or list on the Dashboard that you can add, move, resize and remove.
Snapshot	The daily record of file count, free space and missing episodes that the trend cards draw.
Match	The link between a series and its TVmaze or AniList entry. You can choose or lock it by hand.
Collect	A per-series choice of which episodes count as missing: everything, from an episode onward, or nothing.
Dry run	A rename batch that records what would happen and changes nothing.
Invite	A link, with a QR code, that lets one person use the family portal. Only its fingerprint is stored.
Family portal	A separate listener with its own short list of operations, safe to publish on its own.
Posture check	A coloured tile on the Security page: green is good, amber an optional improvement, red needs attention.

Word	Meaning
Share helper	The root-owned helper on the Pi that mounts network shares for the web service, which has no root of its own.
Funnel	A Tailscale feature that publishes one local port at a public address with a real certificate.
Caddy	A small web server used as a reverse proxy, giving each service on the Pi a clean name and certificate.
mDNS	The name system behind <code>medialedger.local</code> . It needs no router setup.